

Enalapril mitigates senescence and aging-related phenotypes by targeting antioxidative genes via phosphorylated Smad1/5/9

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eLife Assessment

This study provides **valuable** insights into the anti-senescence effects of enalapril, identifying pSmad1/5/9 signaling and associated antioxidant pathways as key mediators of its physiological benefits in aged mice. The authors present **solid** experimental evidence across both in vitro and in vivo systems, demonstrating improved organ function and reduced senescence markers following treatment. Overall, the work supports the repurposing potential of enalapril in aging research and expands understanding of its molecular targets.

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Abstract

Aging increases the risk of a myriad of chronic diseases, which are expensive and difficult to treat owing to their various risk factors. Repurposing existing medications has accelerated the development of therapies aimed at slowing aging. In this study, we revealed that enalapril, a drug widely prescribed for hypertension, can improve both cellular senescence and individual health. Mechanistically, phosphorylated Smad1/5/9 act as pivotal mediators of the anti-senescence properties of enalapril. It stimulates downstream genes involved in cell cycle regulation and antioxidative defense, facilitating cell proliferation and diminishing the production of reactive oxygen species (ROS), thus increasing the antioxidative ability of enalapril. At the organismal level, enalapril has been shown to bolster the physiological performance of various organs; it notably enhances memory capacity and renal function and relieves lipid accumulation. Our work highlights the potential of enalapril to augment antioxidant defenses and combat the effects of aging, thereby indicating its promise as a treatment strategy for aging-associated diseases and its use for healthy aging.

Introduction

Aging is a multifaceted process influenced by numerous factors, and its intricate mechanisms are not yet fully understood. Geroscience is a burgeoning field of study in which researchers hypothesize that aging is a significant risk factor for many chronic diseases. These findings suggest that interventions targeting the mechanisms of aging might delay the onset or slow the progression of these diseases(Kirkland, 2016 [↗](#); Lopez-Otin, Blasco, Partridge, Serrano, & Kroemer, 2023 [↗](#)). Cellular senescence, a pivotal indicator of aging, is characterized by irreversible cell cycle arrest triggered by cellular damage or stress(Khosla, Farr, Tchkonina, & Kirkland, 2020 [↗](#)). During cellular senescence, cells exhibit increased activity of the lysosomal enzyme senescence-associated β -galactosidase (SA- β -gal), decreased proliferation marker Ki67 signals and increased senescence-associated secretory phenotype (SASP) secretion(Coppe et al., 2008 [↗](#); Dimri et al., 1995 [↗](#)).

Senescence is the result of a complex interplay of endogenous and exogenous signaling pathways(Cohen et al., 2022 [↗](#)). Currently, the activation of tumor suppressor pathways, such as p16^{INK4a}/RB, is deemed essential for senescence-related cessation of proliferation(Baker et al., 2011 [↗](#)). The disruption of signaling pathways disrupts tissue homeostasis, leading to proinflammatory responses that are increasingly recognized as fundamental aspects of senescence(Coppe et al., 2008 [↗](#); Shah et al., 2013 [↗](#)). Researchers have suggested that the expression of p16 is suppressed by inhibitor of DNA-binding (ID) proteins. The expression of the *ID1* gene is downregulated by transforming growth factor- β (TGF- β), while it is upregulated by BMP-mediated phosphorylated SMAD signaling in various cell types(Hayashi et al., 2016 [↗](#); Liang, Brunicardi, & Lin, 2009 [↗](#)). SMADs are crucial transcriptional effector factors responsible for transmitting TGF- β /BMP signals(Urist, 1965 [↗](#)). They convey signals from the TGF- β superfamily and homomeric serine/threonine kinases on the cell surface to the nucleus, directly regulating transcriptional programs critical for key events(Miyazono, Maeda, & Imamura, 2005 [↗](#)). Numerous studies have shown that the BMP-SMAD signaling pathway can modulate genes involved in the cell cycle, which affects the balance between cell proliferation and senescence(Genander et al., 2014 [↗](#); Hayashi et al., 2016 [↗](#)). Phosphorylated SMAD plays a crucial role in the induction and maintenance of pluripotency, helping maintain tissue homeostasis during aging(Orlowski, 2017 [↗](#)). However, dysregulation of BMP signaling during aging can lead to fibrosis, impaired wound healing, or excessive tissue calcification, which are detrimental to tissue function(Lyu et al., 2018 [↗](#); Rahman, Akhtar, Jamil, Banik, & Asaduzzaman, 2015 [↗](#)).

Owing to increasing insights into cellular and individual aging, novel aging mechanisms and signaling pathways have been identified, leading to the development of new therapeutic interventions. Senescent cells have recently been targeted in therapy given their substantial impact on aging (Childs et al., 2017 [↗](#)). Consequently, eliminating senescent cells through pharmacological interventions, known as “senolytics”, may delay the onset or slow the progression of age-related diseases and enhance physiological function(Childs et al., 2017 [↗](#)). Research has proven that specific senolytic drugs, such as dasatinib and quercetin, can eliminate senescent cells, reverse aging phenotypes and increase the survival rate of mice(Xu et al., 2018 [↗](#); Zhu et al., 2015 [↗](#)). However, the current senolytics are not universally effective across different cell types and display inconsistent sensitivities, thereby limiting their broad application(Yousefzadeh et al., 2018 [↗](#); Zhu et al., 2016 [↗](#)). In addition to senolytics, researchers have explored pharmacological interventions that aim to ameliorate or reverse aging phenotypes without the need to eliminate senescent cells. This included both the repurposing of existing medications and the development of new ones(Bharath et al., 2020 [↗](#); Harrison et al., 2014 [↗](#)). However, repurposing is particularly attractive due to its cost-effectiveness; these drugs have already undergone rigorous safety evaluations, allowing for quicker implementation in new contexts. For example, drugs such as metformin, which is typically used to treat type 2 diabetes,

and rapamycin, an immunosuppressant, have shown promise in reducing aging phenotypes and extending the lifespan (Bharath et al., 2020 [↗](#); Harrison et al., 2009 [↗](#)). This strategy is an innovative pharmacological intervention in which existing compounds target the aging process.

In this study, we employed a drug repurposing strategy to evaluate the anti-senescence potential of several FDA-approved drugs and identified enalapril as a promising candidate. Our investigation revealed that enalapril increases pSmad1/5/9 levels, which in turn upregulates key downstream genes involved in cell cycle regulation and antioxidative defense, thereby improving cell proliferation, reducing inflammation, and decreasing reactive oxygen species (ROS) levels. Our findings confirmed the anti-senescence mechanism of enalapril at both the cellular and organismal levels. Enalapril has the ability to ameliorate aging-related phenotypes of various organs, with notable effects on the brain, kidney, and liver. Furthermore, treating aged mice with enalapril improved spatial memory and anxiety behavior, alleviated lipid accumulation, and mitigated kidney aging-related phenotypes. Taken together, the potential of enalapril to extend the healthspan and ameliorate age-related diseases makes it a compelling candidate for research and development in the field of anti-aging therapeutics.

Materials and methods

Cell culture

IMR90 cells were cultured in DMEM (Gibco, C11995500) supplemented with 10% fetal bovine serum (Cellmax, SA111) and 1% non-essential amino acids (Gibco, 11140050) at 37°C and 5% CO₂. When the cell density in the in vitro culture reached 80–90% (usually within 2 or 3 days), cell passage was performed.

Drug preparation and addition

Enalapril (MCE, HY-B0331) was dissolved in DMSO to prepare a 10 mM stock solution. When used, it was diluted to 10 μM in the culture medium. After the cells adhered to the surface, enalapril was added for treatment, and RNA and protein were collected for analysis. Recombinant BMP4 protein (MCE, HY-P7007) was dissolved in sterile ddH₂O to prepare a 100 μg/mL stock solution. When used, it was diluted to 50 ng/mL in the culture medium and added to the cells. The BMP receptor inhibitor (LDN193189, MCE, HY-12071) was dissolved in sterile ddH₂O to prepare a 1 mM stock solution. When used, it was diluted to 1 μM or 0.5 μM in the culture medium. The ID inhibitor (Pimozide, MCE, HY-12987) was dissolved in DMSO to prepare a 10 mM stock solution. When used, it was diluted to 5 μM in the culture medium. After the cells adhered to the surface, drugs were added for a 3-day treatment, and samples were collected for analysis.

Senescence-associated β-galactosidase (SA-β-gal) Staining

SA-β-gal staining was performed using a Senescence β-Galactosidase Staining Kit (Sigma-Aldrich, CAS0030). Briefly, 0.6×10^5 cells were seeded in a 6-well plate. The cells were fixed with fixation buffer and incubated at room temperature for 10 minutes. After the cells were washed three times with PBS, the prepared staining mixture was added to the 6-well plate, and the plate was incubated at 37°C overnight. The stained cells were observed and photographed under a microscope.

Western blot

Proteins were extracted using TRIzol reagent (Invitrogen, 15596018CN) and dissolved in 1% SDS, after which SDS-PAGE was performed. The samples were transferred to nitrocellulose membranes and incubated with 5% skimmed milk at room temperature for 30 minutes. The primary antibodies were incubated overnight at 4°C, followed by three washes with PBST. The membranes

were then incubated with secondary antibody at room temperature for two hours. The band signals were visualized using an Odyssey Infrared Imaging System (Odyssey, LI-COR). All antibody information is provided in **Table S4** [↗](#).

RT-qPCR

Total RNA from cultured cells was extracted using TRIzol reagent (Invitrogen, 15596018CN) according to the manufacturer's instructions. The DEPC-treated water-dissolved RNA was subsequently reverse transcribed into cDNA using the HiScript III All-in-one RT SuperMix Perfect for qPCR (Vazyme, R333) following the product manual. The cDNA levels of specific primers were analyzed using the LightCycler[®] 96 qPCR system (Roche), and the results were normalized to those of β -actin. The primers used in this analysis are listed in **Table S1** [↗](#).

RNA-seq and data processing

Total RNA was extracted using TRIzol reagent (Invitrogen, 15596018CN) following the manufacturer's instructions. The extracted total RNA was then sent to Novogene for subsequent cDNA library construction. The library was sequenced with paired-end reads on the Illumina HiSeq PE150 platform.

For data processing, FastQC (version 0.11.5) software was first used for data quality control. Raw paired-end sequencing reads were trimmed by Trim Galore (version 0.6.10) software to remove adapter sequences and low-quality reads. The high-quality RNA-seq reads were mapped to the hg38/mm10 genome using HISAT2 (version 2.2.1) software(Kim, Paggi, Park, Bennett, & Salzberg, 2019 [↗](#)). SAMtools (version 1.17) software was then used to convert the generated SAM files to bam files(Li et al., 2009 [↗](#)). To qualify gene expression, the number of reads mapped to each gene was counted using featureCounts (version 2.0.1) software(Liao, Smyth, & Shi, 2014 [↗](#)). Differentially expressed genes (DEGs) were calculated by R package DESeq2 (version 1.34.0)(Love, Huber, & Anders, 2014 [↗](#)) with the cutoff values of Benjamini-Hochberg adjusted p-value < 0.05 and $|\log_2(\text{fold change})| > 0.5$. The gene expression count matrix was converted into the reads per kilobase per million mapped reads (RPKM) matrix, followed by Z score normalization for visualization.

CUT&Tag and data processing

For the CUT&Tag experiments, 8×10^5 cells were processed using the CUT&Tag Kit (Vazyme, TD903). The cells were collected and incubated with ConA beads and then mixed with the primary antibody pSmad1/5/9 (CST, #13820) overnight at 4°C. After incubation with the secondary antibody for one hour, the cells were further incubated with pA/G-Tnp for one hour. The transposase was then activated for DNA fragmentation. Following DNA extraction, library amplification and purification were performed. The libraries were sequenced using paired-end reads on the Illumina HiSeq PE150 platform.

For data processing, raw CUT&Tag data was trimmed using Trim Galore (version 0.6.10) software, and the clean reads were mapped to the human reference genome hg38 using Bowtie2 (version 2.5.1) software(Langmead & Salzberg, 2012 [↗](#)) with the following parameters: “--very-sensitive --no-mixed --no-discordant -I 10 -X 1000”. The SAM files were transformed into bam files via SAMtools (version 1.17) software(Li et al., 2009 [↗](#)). The CUT&Tag peaks were performed using the MACS2 (version 2.2.8) with the default parameters(Zhang et al., 2008 [↗](#)). Differential peaks were calculated using the R package DiffBind (version 3.4.11) with the cutoff values of adjusted p-value < 0.05 and $|\log_2(\text{fold change})| > 0.5$ and annotated using the R package ChIPpeakAnno (version 3.28.0)(Lihua, Claude, & Nathan, 2010 [↗](#)). To visualize the CUT&Tag signal, the reads per kilobase per million mapped reads (RPKM) were calculated in each 50-bp bin size using DeepTools (version 3.5.2) software(Ramirez, Dundar, Diehl, Gruning, & Manke, 2014 [↗](#)), and the RPKM matrix was normalized using the Z score algorithm.

ChIP-qPCR

At room temperature, approximately 2×10^6 IMR90 cells were cross-linked with 1% paraformaldehyde for 10 minutes. The cross-linking reaction was stopped by adding 0.125 M glycine for 6 minutes. The cells were then harvested and lysed on ice in nuclear lysis buffer (50 mM Tris-Cl, 10 mM EDTA, 1% SDS, and protease inhibitors). Chromatin immunoprecipitation (ChIP) was performed using a Bioruptor (Diagenode) to shear the chromatin. Fragmented chromatin (25 μ g) was immunoprecipitated with 5 μ g of anti-pSmad1/5/9 antibody (CST, #13820) or anti-rabbit IgG (Abcam, ab171870). The immunoprecipitated DNA was quantified via RT-qPCR. The amount of DNA in the immunoprecipitates was calculated as a percentage relative to 10% of the input chromatin. All ChIP-qPCR-related primers are available in [Table S2](#).

Immunofluorescence

The slides were washed twice in PBS, fixed with 4% PFA for 10 minutes, and washed again three times with PBS. Subsequently, permeabilization was performed via incubation with 0.5% Triton X-100 for 10 minutes, followed by three additional washes with PBS. Next, blocking was performed with 1% BSA, and after blocking, the slides were washed three times with PBS. The primary antibody (Ki67, Abcam, ab15580) was then diluted at a ratio of 1:500 in PBST (PBS + 0.1% Tween 20) and incubated overnight at 4°C. Afterward, the slides were washed four times with PBST, with each wash lasting 5 minutes. The secondary antibody was diluted at a ratio of 1:500 in PBST (1 $\times$ PBS + 0.1% Triton X-100) and incubated at room temperature for 1–1.5 hours. After three washes with PBST, a final wash with PBS was conducted. DAPI was diluted to a final concentration of 1–2 ng/ μ L and incubated at room temperature for 3–5 minutes. Following three washes with PBS and one wash with ddH₂O, the slides were sealed with Fluoromount-G[®] (SouthernBiotech, 0100-010) and covered with a coverslip.

Lentiviral production and viral transduction

First, the pLKO.1 empty vector or vectors encoding shRNAs targeting *ID1* or *ID2* ([Table S3](#)), along with the assistant plasmids psPAX2 and pMD2.G, were cotransfected into HEK293T cells. After 18 h of transfection, the medium was replaced with DMEM supplemented with 30% complement-inactivated FBS, and viral supernatants were collected after 24 h and 48 h of transfection. When IMR90 cells reached 70% confluence, they were infected with the packaged virus with 5 μ g/mL polybrene (MCE, HY-K1057). At 36–48 hours after infection, successfully transduced cells were selected via treatment with 1 μ g/mL puromycin (M&C Gene Technology, MA009) for 24–36 hours.

Detection of Reactive Oxygen Species (ROS)

IMR90 cells were harvested from a 6-well plate. After the cells were washed, the ROS-sensitive fluorescent probe 2',7'-dichlorofluorescein diacetate (DCFH-DA, MCE, HY-D0940) was used for detection. A 10 mM stock solution of DCFH-DA was prepared by dissolving it in anhydrous DMSO. This stock solution was subsequently diluted in serum-free DMEM to obtain a 10 μ M working solution of DCFH-DA. The samples were incubated with the DCFH-DA probe at 37°C for 30 minutes. After incubation, the samples were washed to remove excess probe. The fluorescence intensity was measured via a fluorescence microscope. The fluorescence intensity corresponds to the level of ROS present in the sample.

Mice feeding

All the mouse experiments were approved by the Institutional Animal Care and Use Committee of Peking University. 12-month-old C57BL/6 mice (purchased from Wukong Biotech Co., Ltd.) were housed in a specific pathogen-free (SPF) animal facility under controlled environmental conditions with a temperature range of 20–25°C, 55 $\pm$ 10% humidity, and a 12 h light/dark cycle. The

mice had ad libitum access to food and water. A daily dosage of 30 mg/kg of enalapril was dissolved into the mice's drinking water. In the LDN treatment group, mice received intraperitoneal injections of the BMP receptor inhibitor LDN193189 (3 mg/kg, MCE, HY-12071) every other day. Four to ten mice were included in each group.

Forelimb grip strength test

The mice were placed on a grip strength meter, and their tails were gently pulled to allow them to grasp the pull bar with their forelimbs. The grip strength meter readings were recorded at the point of maximum force exerted by each mouse. The measurements were repeated multiple times to ensure reliable and consistent results.

Rotarod fatigue test

The mouse was placed on the rotarod, and rotation was initiated. The device was set to an accelerating mode that increased from 4 to 40 revolutions per minute over 300 seconds. The time at which each mouse fell off or lost balance was recorded. The test was repeated multiple times to assess the endurance and motor performance of the mice.

Y-maze test

The Y-maze consists of three opaque arms arranged at 120° angles relative to each other. The mice were placed in the central position of the maze and allowed to explore the three arms freely for 10 minutes. Each consecutive entry into one of the three arms was defined as spontaneous alternation behavior. The total number of arm entries and the number of spontaneous alternations were recorded. The percentage of spontaneous alternation behavior was then calculated.

Open field test

The mice were removed from their cages and placed in an open field, with the central area designated. The mice were allowed to move freely within the open field for 10 minutes, and the number of times that they crossed the central area was recorded.

Mouse serum extraction and analysis

After the mice were anesthetized, blood (approximately 1 mL per mouse) was collected from the eyeballs. The blood was allowed to clot and settle at room temperature. The mixture was then centrifuged at 3,000 rpm for 20 minutes, and the upper clear layer, which was the serum, was collected. The serum was subsequently subjected to biochemical assays, including ALT, CREA, LDH, TG, LDL, and HDL tests.

Mouse tissue extraction

After the mice were euthanized, they were perfused with prechilled PBS until no blood was expelled. The heart, liver, spleen, kidney, brain, and muscle tissues were harvested. A portion of each tissue sample was used for RNA and protein extraction for sequencing and Western blot analysis, respectively, while another portion was fixed in 4% paraformaldehyde (PFA). The tissues were then subjected to either frozen or paraffin embedding. Various staining techniques, including β -galactosidase (β -gal) staining, periodic acid-Schiff (PAS) staining, Congo red staining, Sirius red staining, Oil Red O staining, and hematoxylin and eosin (H&E) staining, were performed.

Pathway enrichment analysis

Gene Ontology (GO) enrichment analysis was performed with Metascape(Zhou et al., 2019 [DOI](#)), and gene set enrichment analysis (GSEA) was performed with the R package ClusterProfiler (version 4.2.2)(Wu et al., 2021 [DOI](#)). Pathways with p values < 0.05 were considered significantly enriched.

Representative terms were visualized with the ggplot2 (version 3.3.6) R package and ComplexHeatmap (version 2.10.0) R package (Gu, 2022 [↗](#)).

Statistical analysis

The data are expressed as means $\pm$ SEMs. The data were analyzed via a two-tailed unpaired Student's *t* test. A *p* value < 0.05 was considered statistically significant.

Results

Enalapril restrains cellular replicative senescence

To identify drugs that are capable of delaying cellular senescence, we established an in vitro replicative senescence model using primary human embryonic lung fibroblasts (IMR90). Compared with young cells, long-term-cultured IMR90 cells presented a characteristic senescence phenotype, characterized by increased SA- β -gal staining, increased expression of the cell cycle arrest factors p16 and p21, and reduced expression of the proliferation marker Ki67 (Fig. S1A–C [↗](#)). Additionally, the release of SASP factors that significantly alter the cellular microenvironment, including the well-described factors *IL-1 β* , *IL-6*, and *CXCL10*, has also been reported to increase the degree of senescence (Fig. S1D [↗](#)).

Employing a drug repurposing strategy, we discovered that enalapril, an antihypertensive medication, delayed senescence. Notably, enalapril treatment at various concentrations reduced SA- β -gal staining and increased Ki67 positivity in IMR90 cells (Fig. 1A–B, D–E [↗](#)). Western blot analysis confirmed a decrease in the senescence markers p16 and p21 with the optimal effect of 10 μ M enalapril (Fig. 1C [↗](#)). To further analyze the effects of enalapril on transcriptome dynamics upon senescence entry, RNA-seq was performed across enalapril-treated cells at different concentrations. In general, we found that the expression of SASP genes decreased significantly (Fig. 1F [↗](#)). The expression of cell cycle arrest factors typically increased during senescence; however, the expression of these factors decreased after exposure to enalapril, whereas the expression of cell cycle genes related to cell proliferation increased upon treatment with enalapril at the optimal concentration of 10 μ M (Fig. 1F [↗](#)). We therefore treated the cells with 10 μ M enalapril via RT-qPCR and detected that many prominent SASP genes, such as *IL-1 β* and *IL-6*, were significantly reduced compared with those in the control (Fig. 1G [↗](#)). Moreover, enalapril treatment increased the cell growth rate and delayed the transition to the growth arrest phase (Fig. 1H [↗](#)). Our comprehensive analysis underscores enalapril's potential as an anti-senescence agent, warranting further exploration for its role in combating cellular senescence.

The function of enalapril depends on phosphorylated Smad1/5/9

Given the potential of enalapril to decelerate senescence, we screened several important targets of the senescence-related pathway to identify the underlying mechanism of senescence entry. We found that enalapril notably elevated the phosphorylation level of Smad1/5/9 without affecting Smad2/3 phosphorylation, effectively reversing the decline of pSmad1/5/9 levels in senescent cells, suggesting a possible role for the BMP-pSmad1/5/9 pathway (Fig. 2A [↗](#), Fig. S2A, B [↗](#)). To validate the pathway alterations in the gene expression profile, GSEA confirmed that enalapril treatment increased BMP signaling (Fig. 2B [↗](#)). Concurrently, the level of the Smad4 protein, which is required for pSmad1/5/9 to enter the nucleus, also increased, suggesting that enalapril might facilitate the nuclear entry of pSmad1/5/9 and modulate downstream genes (Fig. 2A [↗](#)) (Guglielmi et al., 2021 [↗](#); Senft, Bikoff, Robertson, & Costello, 2019 [↗](#)). Notably, the expression of Smad4 was reduced in senescent cells, supporting the notion that enalapril reverses senescence-associated

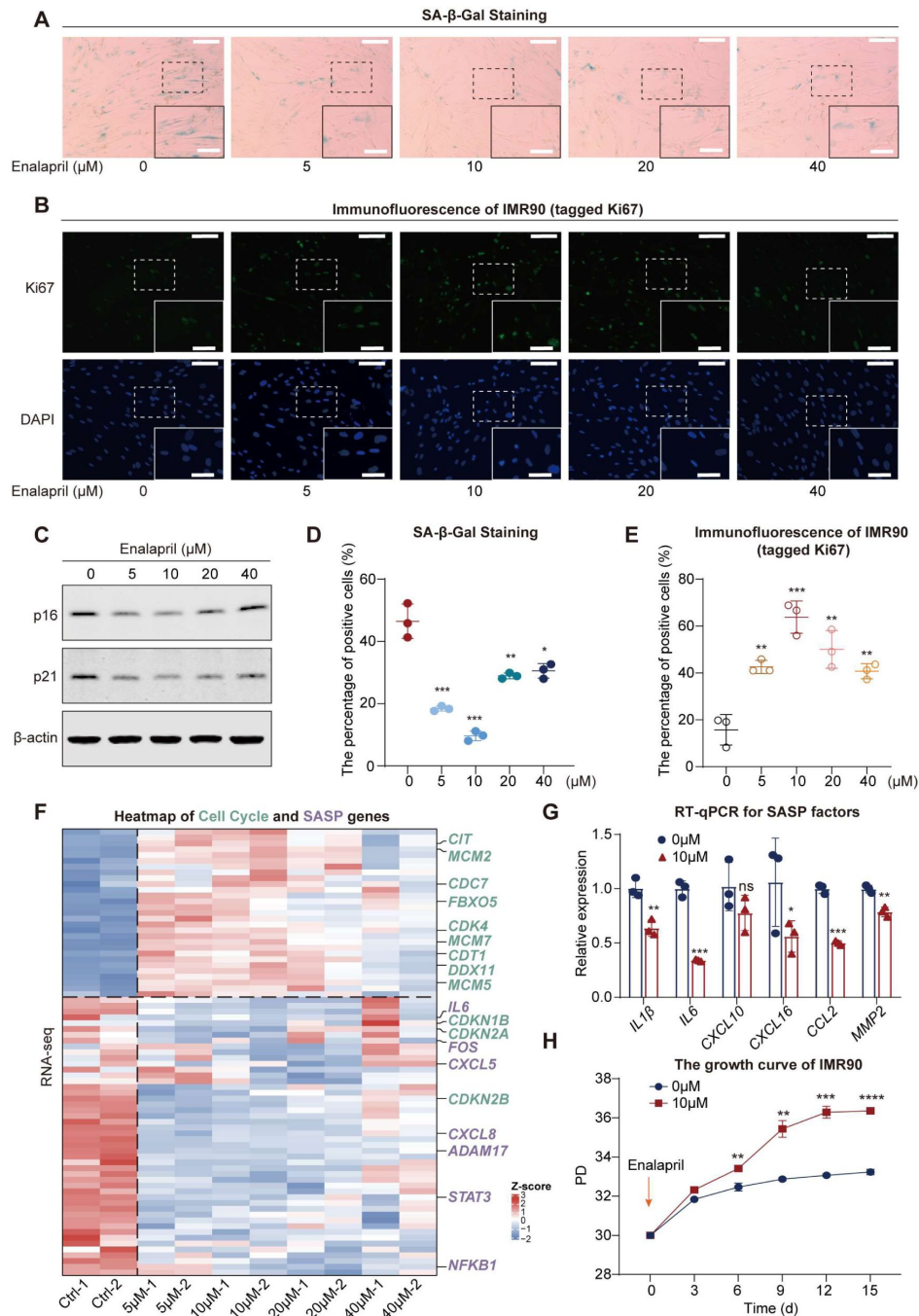


Figure 1

Enalapril alleviates cellular senescence.

(A, D) SA-β-Gal staining (A) and statistical analysis of SA-β-Gal ratios (D) in IMR90 cells after enalapril treatment. Scale bars, 200 μm. Enlarged scale bars, 100 μm. (B, E) Ki67 immunofluorescence experiment (B) and statistical analysis of Ki67 intensity (E) after enalapril treatment. Scale bars, 80 μm. Enlarged scale bars, 40 μm. (C) Western blot analysis showing the protein levels of p16 and p21 in IMR90 cells after enalapril treatment. (F) Heatmap showing cell cycle-related gene (green) upregulation and SASP gene (purple) downregulation after enalapril treatment. (G) RNA expression of SASP factors in IMR90 cells after enalapril treatment, with blue bars representing the control group and red bars representing the enalapril-treated group. (H) Growth curves of IMR90 cells after enalapril treatment. The x-axis represents the passage time, while the y-axis represents population doubling (PD). A two-tailed t test was employed; ns indicates no significant difference, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

defects in BMP signaling (**Fig. S2B**). To further assess the impact on gene regulation, we employed CUT&Tag analysis of pSmad1/5/9 and found that enalapril treatment increased the pSmad1/5/9 binding signals at the transcription start site (TSS) regions (**Fig. 2C**).

To explore the potential role of pSmad1/5/9 in anti-senescence mechanisms, we experimentally manipulated its levels by supplementing with BMP4 to increase pSmad1/5/9 levels, or by reducing it using the BMP receptor inhibitor LDN193189 (LDN) and shRNA-mediated knockdown of the BMP receptor BMPR1A. Our findings revealed that BMP4 increased pSmad5 levels, which correlated with decreased p16 and p21, reduced β -gal staining and increased Ki67 positivity, indicating a mitigated senescent phenotype (**Fig. S3A–C**). In contrast, when cells were treated with the BMP receptor inhibitor LDN or subjected to *BMPR1A* knockdown, the changes in pSmad1/5/9 levels and senescence markers were completely reversed, highlighting that the suppression of pSmad1/5/9 accelerates the progression of senescence (**Fig. S4A–G**).

Encouraged by the observed association between pSmad1/5/9 and senescence, we tested whether pSmad1/5/9 mediates the anti-senescence effects of enalapril by co-treating with enalapril and either LDN or shRNA targeting *BMPR1A*. RNA-seq analysis revealed that LDN upregulated SASP genes and downregulated cell cycle genes. On the basis of these findings, the subsequent addition of enalapril did not reverse these changes, which suggests that the anti-senescence function of enalapril relies on pSmad1/5/9 (**Fig. 2D**). Furthermore, western blot results also revealed that in the presence of LDN, neither enalapril nor BMP4 restored Smad1/5/9 phosphorylation or the expression of p16 and p21 (**Fig. 2E**). Similarly, β -gal and Ki67 staining also revealed that enalapril failed to counteract the senescence-promoting effects induced by either LDN treatment or *BMPR1A* knockdown (**Fig. 2F, G**, **Fig. S4F, G**). Collectively, these results indicate that elevated pSmad1/5/9 could increase the expression of critical genes, thereby exerting a potent anti-senescence effect.

pSmad1/5/9 modulates the cell cycle and SASP secretion by upregulating *ID1*

The ID family, which is recognized as a key downstream target of pSmad1/5/9, serves as a biomarker for Smad1/5/9 phosphorylation levels (Genander et al., 2014; Hayashi et al., 2016; Ramachandran et al., 2018; Sun et al., 2022; Ying, Nichols, Chambers, & Smith, 2003). Coincidentally, we found that *ID1* was the top-ranked transcription factor that pSmad1/5/9 bound to its promoter region via CUT&Tag analysis after enalapril treatment (**Fig. 3A**). Moreover, IGV analysis confirmed that enalapril significantly increased pSmad1/5/9 enrichment at the *ID1* promoter (**Fig. 3B**). Simultaneously, the expression levels of both *ID1* and *ID2* were elevated (**Fig. 3C, D**). Conversely, inhibiting pSmad1/5/9 with LDN or shRNA targeting *BMPR1A* led to decreased *ID1* and *ID2* levels, confirming the regulatory link between pSmad1/5/9 and the ID family (**Fig. 3E**, **Fig. S4D**). This finding indicated that the ID family was indeed a downstream gene regulated by pSmad1/5/9, and the increase in *ID1* induced by enalapril further confirmed that enalapril could upregulate pSmad1/5/9 levels, thereby affecting the expression of downstream genes.

Prior studies have indicated that the ID family can suppress p16 and p21 expression, thus inhibiting cell cycle arrest and delaying cellular senescence (Hayashi et al., 2016; Lyden et al., 1999; Ying et al., 2003). This prompted us to examine ID protein expression during senescence, where we observed a decrease in *ID1* and *ID2* levels (**Fig. S5A**). The knockdown of *ID1* and *ID2* via shRNA or an *ID1* inhibitor resulted in a decreasing trend of *ID1* or *ID2* expression and a significant increase in p16 and p21 expression (**Fig. 3F, I**). In addition, while β -gal increased the proportion of positively stained cells, the Ki67 proliferation signal decreased nearly fourfold after the knockdown of *ID1* or *ID2*, resembling the changes observed during senescence (**Fig. S5B, C**). Given that enalapril reduced SASP factors while LDN increased them, we investigated whether *ID1*, which acts as a transcriptional repressor downstream of pSmad1/5/9,

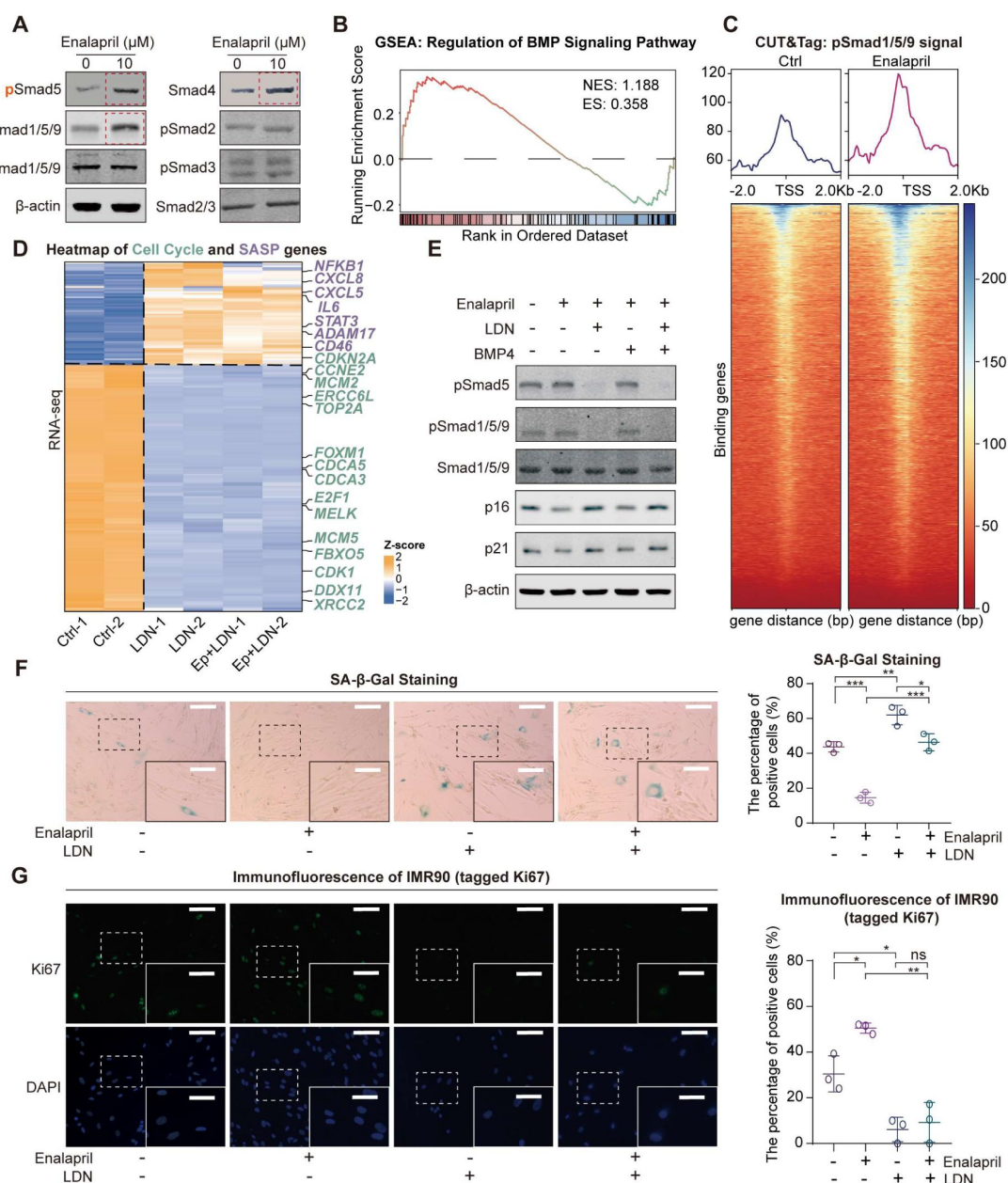


Figure 2

Phosphorylated Smad1/5/9 mediate the anti-senescence effect of enalapril.

(A) Western blot results for the detection of pSmad5, pSmad1/5/9 and other proteins after enalapril treatment. (B) GSEA showing the enrichment of the BMP signaling pathway after enalapril treatment. NES, normalized enrichment score. ES, enrichment score. (C) Profile of pSmad1/5/9 enrichment at the TSS regions in the control group (Ctrl) and the enalapril-treated group (Enalapril). (D) Heatmap of cell cycle (green) and SASP-related genes (purple) in the control group (Ctrl), BMP receptor inhibitor-treated group (LDN193189, LDN), and enalapril and BMP receptor inhibitor-cotreated group (EP+LDN). (E) Western blot analysis showing the protein levels of pSmad1/5/9, p16, and p21 in response to different combinations of enalapril, a BMP receptor inhibitor (LDN193189, LDN), and BMP4. (F) SA-β-Gal staining (left) and corresponding ratio statistical chart (right) of IMR90 cells treated with different combinations of enalapril, a BMP receptor inhibitor (LDN193189, LDN), and BMP4. Scale bars, 200 μm. Enlarged scale bars, 100 μm. (G) Ki67 immunofluorescence experiment (left) and intensity statistical chart (right) of IMR90 cells treated with different combinations of enalapril, a BMP receptor inhibitor (LDN193189, LDN), and BMP4. Scale bars, 80 μm. Enlarged scale bars, 40 μm. A two-tailed t test was employed; ns indicates no significant difference, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

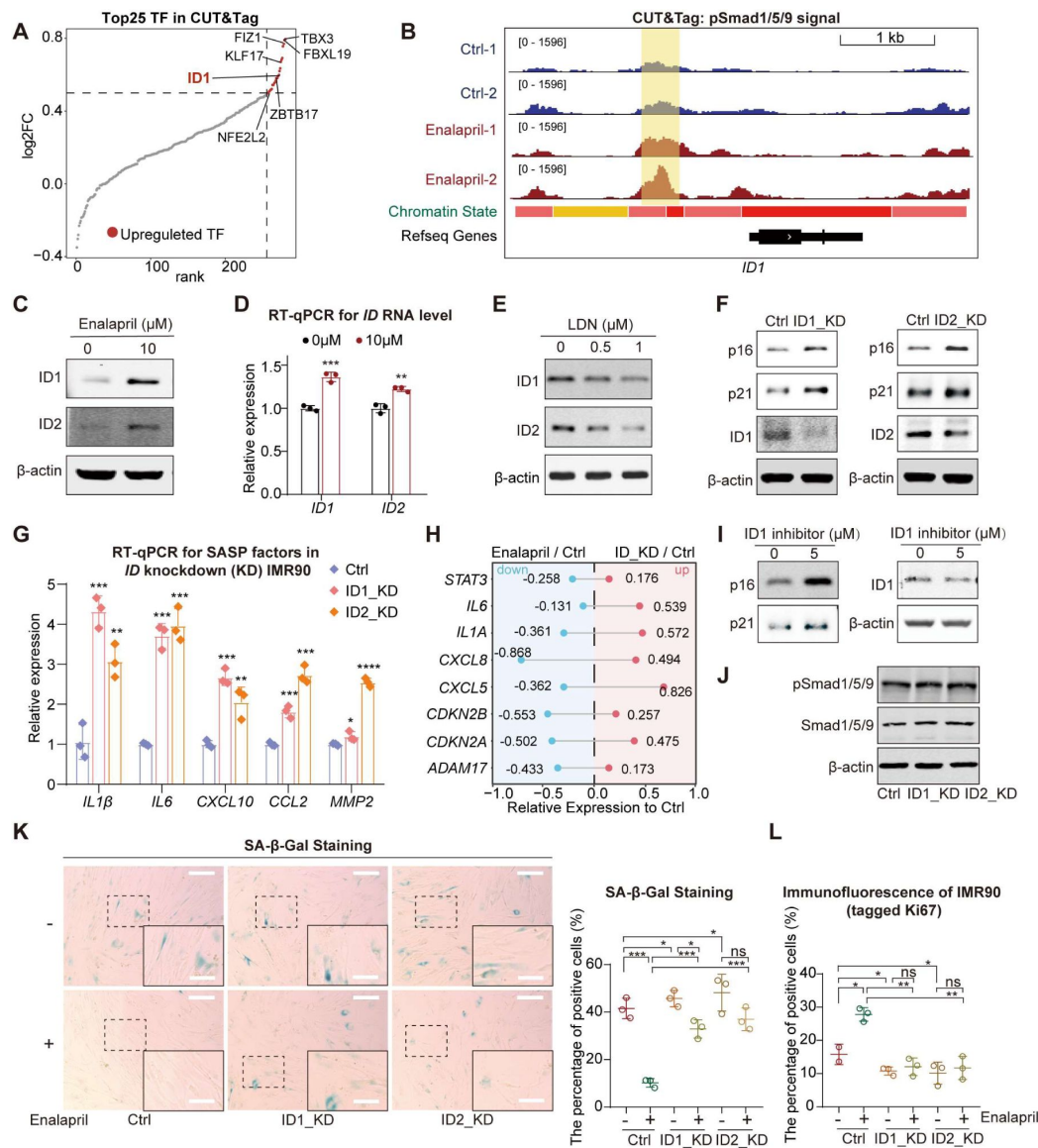


Figure 3

pSmad1/5/9 regulate ID-induced inhibition of p16, p21 and SASP.

(A) Changes in the peaks of all transcription factors identified by CUT&Tag following enalapril treatment, with red dots indicating transcription factors with increased peak intensity (Upregulated TFs). (B) Integrative Genomics Viewer (IGV) showing pSmad1/5/9 signals near the *ID1* region between the control group (Ctrl) and the enalapril treatment group (Enalapril). The vertical yellow boxes indicate regions with increased signal intensity. (C) Western blot analysis showing the changes in the protein levels of ID1 and ID2 following enalapril treatment. (D) Changes in the RNA levels of ID1 and ID2 following enalapril treatment. (E) Western blot analysis showing the protein levels of ID1 and ID2 following treatment with a BMP receptor inhibitor (LDN193189, LDN). (F, I) Western blot showing the changes in the protein levels of p16 and p21 following the knockdown of *ID1* or *ID2* (F) or the inhibition of *ID1* (I). (G) RNA expression of SASP factors after *ID* knockdown, with pink representing *ID1* knockdown and orange representing *ID2* knockdown. (H) Normalized average RNA expression levels of selected SASP factors and cell cycle arrest factors after enalapril treatment and *ID* knockdown. Positive values indicate upregulation, while negative values indicate downregulation. The values represent the expression levels relative to those of the Ctrl. (J) Western blot analysis showing pSmad1/5/9 levels following *ID1* and *ID2* knockdown. (K) SA- β -Gal staining (left) and SA- β -Gal ratio quantification (right) in the control group (Ctrl) and *ID*-knockdown groups (*ID1_KD*, *ID2_KD*) with or without enalapril treatment. Scale bars, 200 μ m. Enlarged scale bars, 100 μ m. (L) Ki67 immunofluorescence intensity in the control group (Ctrl) and *ID* knockdown groups (*ID1_KD*, *ID2_KD*), with or without enalapril treatment. A two-tailed t test was employed, ns indicates no significant difference, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

could modulate SASP expression. Indeed, the expression of traditional SASP factors, such as *IL1 β* and *IL6*, which decrease with enalapril treatment, significantly increased after *ID* knockdown (Fig. 3G). By overlapping the genes downregulated by enalapril with those upregulated upon *ID* knockdown, several SASP genes, as well as *CDKN2A* and *CDKN2B*, were identified (Fig. 3H). Consistent with these findings, we found that SASP factor expression also increased in *ID* knockout mouse bone marrow cells in previously reported data (Fig. S5E)(Fei et al., 2023). Importantly, *ID* knockdown did not affect pSmad1/5/9 levels, confirming that ID is a downstream effector of pSmad1/5/9, influencing the expression of genes such as p16, p21, and SASP-related genes (Fig. 3J).

After validating the prosenescence effects of *ID* through shRNA targeting *ID* or the use of small molecule inhibitors, we next examined whether the anti-senescence effects of enalapril are mediated by ID proteins. The continued addition of enalapril to these *ID*-knockdown cells did not significantly alter β -gal or Ki67 signals compared with those in the *ID*-knockdown group without enalapril, implying that the benefits of enalapril may be ID dependent (Fig. 3K-L, Fig. S5D). Taken together, these findings indicate that enalapril elevates pSmad1/5/9 levels, which in turn upregulate *ID1* expression. This dual action has two key implications: first, it confirms that enalapril can modulate the expression of pSmad1/5/9 downstream genes. Second, by upregulating *ID1*, enalapril can suppress the expression of p16, p21, and SASP-related genes, thereby potentially restoring cell proliferation and reducing the expression of inflammatory factors. These findings suggest that the anti-senescence mechanism of enalapril involves the pSmad1/5/9-ID1 axis, highlighting a novel pathway for its therapeutic effects on aging.

Enalapril has antioxidative effects through pSmad1/5/9

Since an increase in pSmad1/5/9 after enalapril treatment was observed and pSmad1/5/9 is a transcription factor, to understand the in-depth mechanism of enalapril regulation, we further explored the downstream genes of pSmad1/5/9. Analysis of CUT&Tag-upregulated peaks after enalapril treatment revealed that enalapril treatment led to enrichment of pSmad1/5/9 in the GO pathway associated with DNA replication, the cell cycle and the oxidative stress response, all of which are crucial for cell growth (Fig. 4A). Moreover, transcriptome analysis revealed that the genes whose expression increased after enalapril treatment were also enriched in pathways related to the response to oxidative stress (Fig. 4A). Notably, genes upregulated by enalapril in response to oxidative stress included antioxidative enzymes, electron transport chain components and lipid oxidation inhibitors, with a significant increase in pSmad1/5/9 at their TSS regions (Fig. 4B, C). Specifically, we also examined the promoter regions of antioxidative genes, such as the antioxidative enzyme *PRDX5* and the lipid oxidation inhibitor *TXN*, and found increased pSmad1/5/9 enrichment along with upregulated gene expression (Fig. 4D, F-H). During cellular senescence, a decrease in these antioxidative genes exacerbates oxidative stress, reduces energy metabolism, and increases lipid oxidation, leading to the release of inflammatory cytokines (Fig. 4E)(Coppe et al., 2008).

It has been reported that changes in the expression of antioxidative genes can affect the ROS level of cells; thus, we then used DCFH-DA probes to detect ROS levels in cells(Coppe et al., 2008). The results revealed that cellular senescence increased ROS, whereas enalapril slowed the increase in ROS, indicating its ability to upregulate antioxidative genes and reduce ROS, thus affecting cellular senescence (Fig. 4I). Importantly, the continued addition of BMP receptor inhibitors LDN reversed the beneficial effects of enalapril on ROS and decreased pSmad1/5/9 enrichment at the promoters of these antioxidative genes (Fig. 4G, I). Similarly, shRNA-mediated knockdown of *BMPR1A* also led to decreased expression of antioxidative genes such as *PRDX5* and *TXN* (Fig. S4D). Collectively, these results reinforce the notion that enalapril promotes the upregulation of antioxidative genes through pSmad1/5/9, reduces ROS levels and alleviates oxidative stress, thus playing a crucial role in delaying senescence.

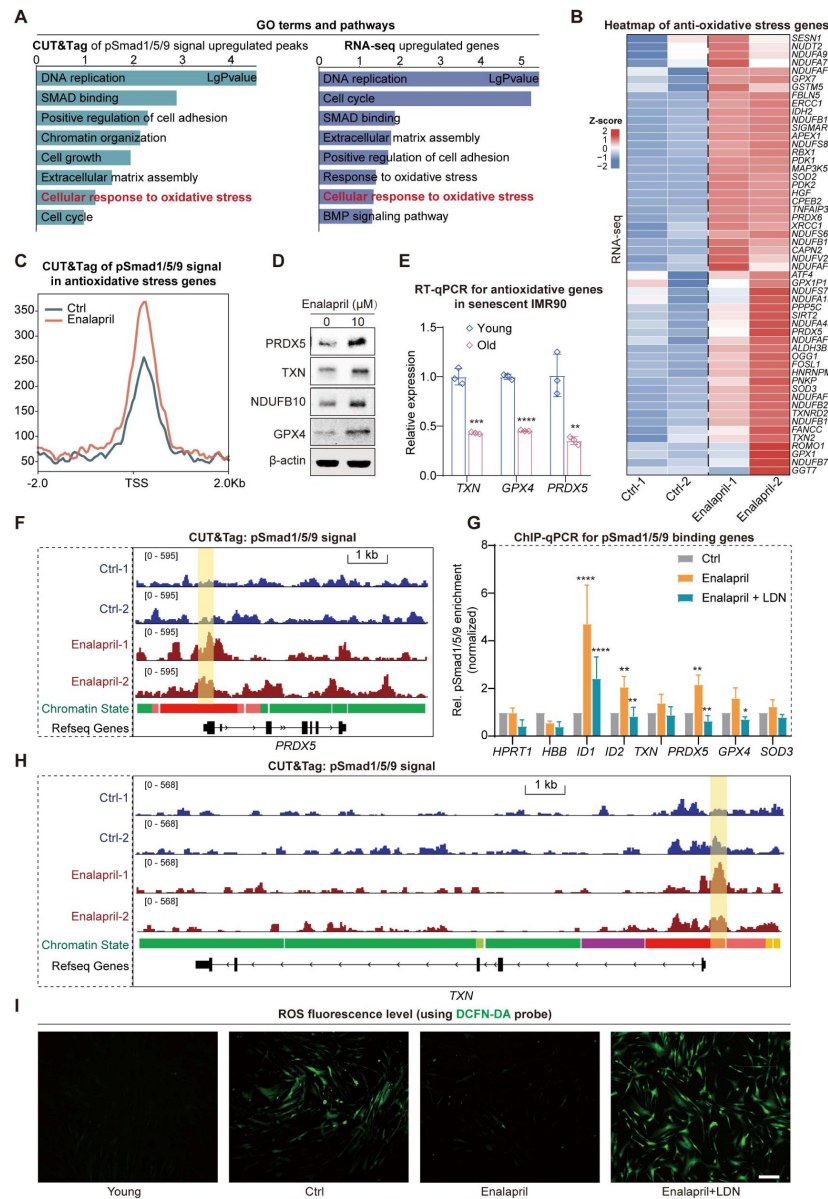


Figure 4

pSmad1/5/9 increase the expression of antioxidative genes.

(A) Bar plot showing the enriched GO terms and pathways associated with the CUT&Tag-upregulated peaks (left) and the upregulated RNA-seq genes (right) following enalapril treatment. (B) Heatmap showing the changes in the RNA expression of antioxidative genes. (C) Profile plot showing the increase in the pSmad1/5/9 binding signal of antioxidative genes in the TSS region following enalapril treatment. (D) Western blot analysis showing the protein levels of representative antioxidative genes after enalapril treatment. (E) Relative RNA levels of *TXN*, *GPX4*, and *PRDX5* during cellular senescence. The blue bars represent young cells, and the red bars represent senescent cells. (F, H) Integrative Genomics Viewer (IGV) showing pSmad1/5/9 signals near the *PRDX5* (F) and *TXN* (H) regions between the control group (Ctrl) and the enalapril treatment group (Enalapril). The vertical yellow boxes indicate regions with increased signal intensity. (G) ChIP-qPCR results showing pSmad1/5/9 levels at many peaks following enalapril or combination treatment with enalapril and LDN193189. The y-axis represents the normalized pSmad1/5/9 signals relative to 10% input. pSmad1/5/9 enrichment for *HPRT1* and *HBB* served as a negative control, and pSmad1/5/9 enrichment for *ID1* and *ID2* served as positive controls, as previously described. A two-tailed t test was employed, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. (I) Detection of ROS fluorescence levels via the DCFH-DA probe in young cells, senescent cells (Ctrl), cells treated with enalapril, and cells treated with a combination of enalapril and LDN193189. Scale bars, 200 μ m.

Enalapril upregulates pSmad1/5/9 and improves physiological function in mice

Enalapril, which is known for its anti-senescence potential, was further investigated for its impact at the organ and individual levels in 12-month-old mice (**Fig. 5A**). After continuous feeding at a dosage of 30 mg/kg/day for 3 months, we conducted a comprehensive RNA-seq analysis of six key organs—brain, kidney, liver, heart, muscle, and spleen—to assess the impact of enalapril on the transcriptome. We found that enalapril induced organ-specific gene expression changes, thereby affecting the physiological functions of these genes. Moreover, GSEA revealed upregulated pathways related to forebrain and kidney development, long-chain fatty acid metabolism and other processes, indicating that enalapril has a beneficial effect on organ functions (**Fig. S6A**). Additionally, enalapril reduced the expression of SASP factors in multiple organs, particularly the CCL, CXCL, and MMP families (**Fig. 5B**). Serum SASP array analysis also revealed a decrease in the secretion of certain SASP proteins, such as CCL6, CCL11, CXCL13, IGFBP-1, and MMP3, suggesting that enalapril has anti-inflammatory effects in mice (**Fig. 5C**).

Given the known upregulation of pSmad1/5/9 levels and antioxidative genes in cellular models by enalapril, we measured Smad1/5/9 phosphorylation and the expression of the antioxidative genes *Txn1*, *Txn2*, and *Prdx5* in the tested organs. We observed increased pSmad1/5/9 levels, especially in the brain, kidney and liver, along with upregulated *Id1* and antioxidative genes, notably *Txn1* and *Txn2* (**Fig. 5D**, **Fig. S6B**). Further analysis of BMP signaling and antioxidative stress-related pathways in these organs revealed increased gene expression after enalapril treatment, indicating the activation of this pathway (**Fig. 5E**). Building on these findings, intraperitoneal injection of the BMP inhibitor LDN reversed the effects of enalapril. Specifically, SASP factor levels in serum, which were reduced by enalapril, were significantly increased. Additionally, the elevated pSmad1/5/9 and antioxidative genes in the brain, kidney and liver were markedly downregulated upon LDN treatment (**Fig. S7A, B**). These findings align with our cellular findings, implying that enalapril enhance the pSmad1/5/9-antioxidative gene axis to increase organ physiological function and delay individual aging.

Enalapril alleviates aging-related behavior in naturally aged mice

Building on our finding of pSmad1/5/9 activation and antioxidative genes upregulation in multiple organs, we further investigated the broader phenotypic effects of enalapril in mice. After three months of enalapril feeding, behavioral experiments, serum tests, and tissue section staining were performed. Notably, enalapril had no significant effect on body weight in the mice, suggesting that the drug has no overt toxic side effects (**Fig. 6A**). However, enalapril improved the decrease in age-related mobility, as evidenced by increased grip strength and extended time in the rotarod fatigue experiments, suggesting increased muscle strength and endurance (**Fig. 6D, E**).

During the aging process, the brain undergoes varying degrees of functional decline. To assess the impact of enalapril on brain function, we conducted Y-maze and open field tests. The Y-maze, which is frequently employed to assess short-term memory in mice by measuring spontaneous alternation as a marker of spatial working memory, indicates notable improvements. Under normal circumstances, mice prefer to explore new maze branches rather than return to previously visited branches. Several brain regions, including the hippocampus, septum, basal forebrain, and prefrontal cortex, contribute to this task, and the age-related decline in spontaneous alternation ability in mice is associated with degeneration or lesions in brain regions such as the hippocampus (Radulescu, Cerar, Haslehurst, Kopanitsa, & Barnes, 2021). Enalapril promoted a substantial increase in the rate of spontaneous alternation in the mice, indicating enhanced spatial memory (**Fig. 6B, C**). As aging progresses, anxiety-like behaviors often increase with age in mice (Radulescu et al., 2021). The open-field test, a valuable tool for assessing autonomous and exploratory behaviors in new environments, demonstrated that enalapril increased the number of central area crossings, suggesting a reduction in aging-related

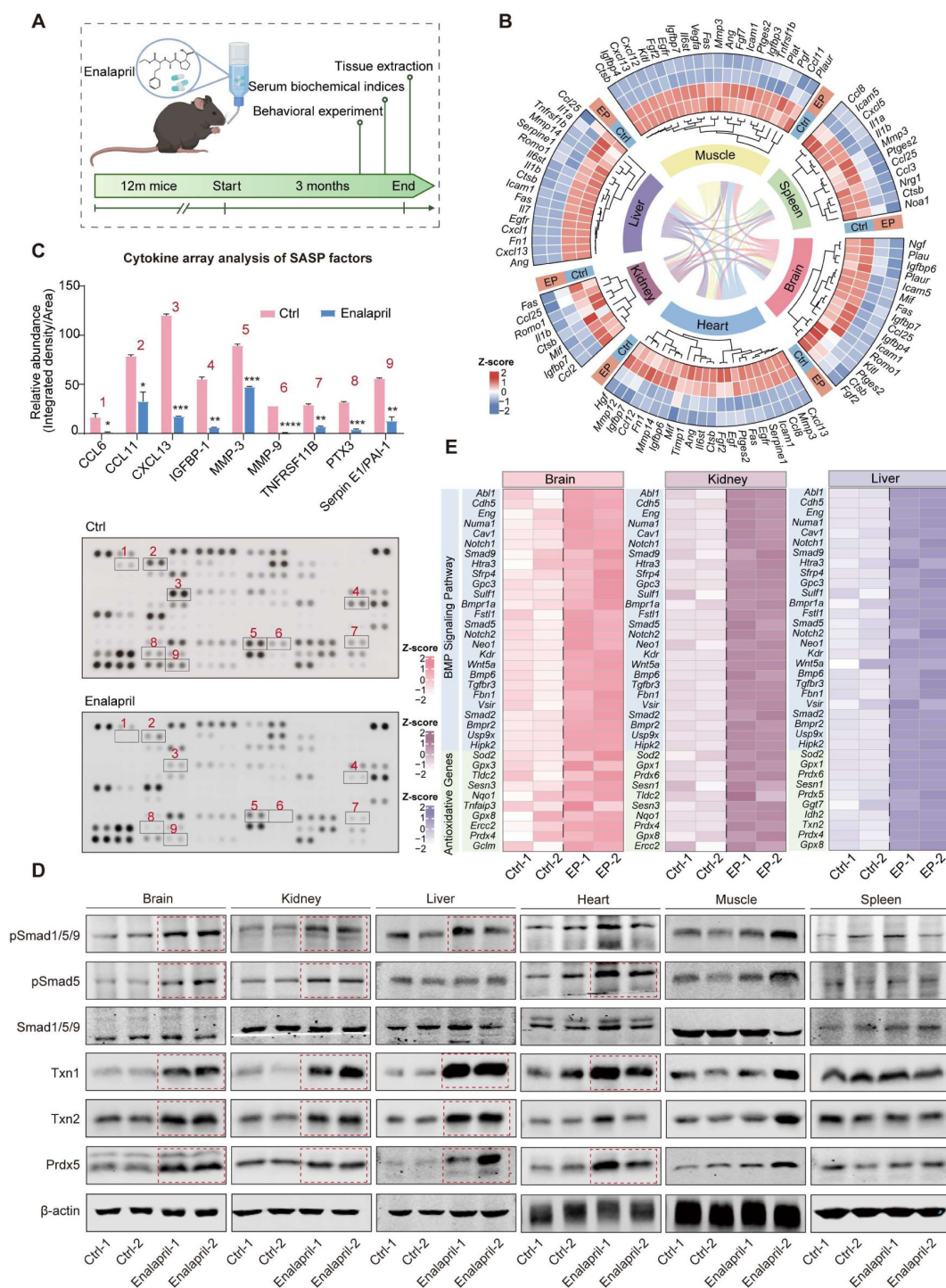


Figure 5

Enalapril exerts its anti-aging effects on mouse organs via pSmad1/5/9-antioxidative genes.

(A) A schematic workflow of the mouse feeding and experimental procedures. Created with [BioRender.com](https://www.biorender.com). (B) Heatmap of RNA levels of SASP factors in various organs following enalapril treatment (EP). (C) Cytokine array analysis of secreted proteins (below) and relative quantitation (above) of SASP factors in control serum (Ctrl) and enalapril-treated serum (Enalapril). (D) Western blot analysis showing the protein levels of pSmad1/5/9 and antioxidative genes in various organs following enalapril treatment. (E) Heatmap of the RNA levels of BMP signaling pathway-related genes and antioxidative genes in the brain, kidney, and liver following enalapril (EP) treatment.

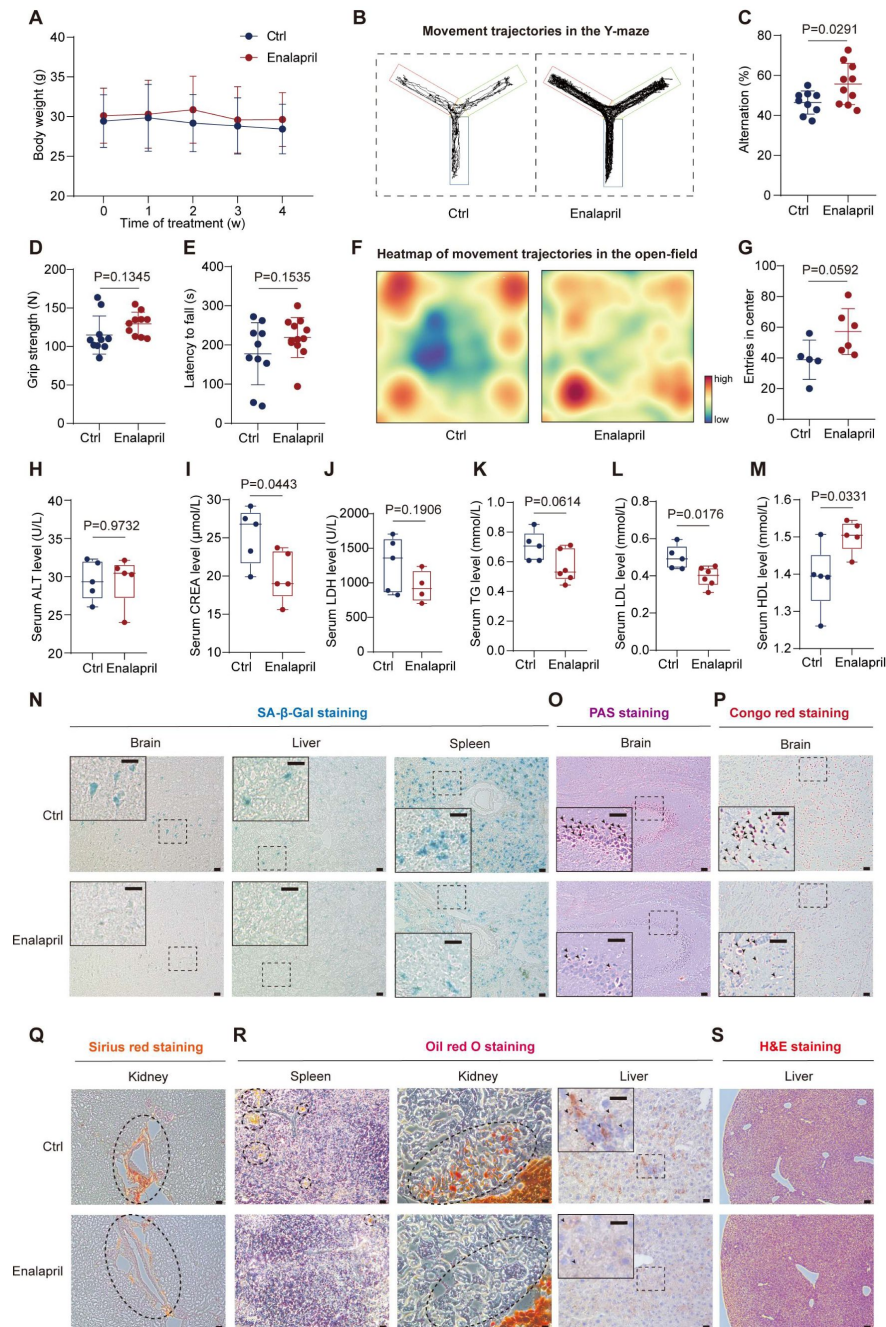


Figure 6

Enalapril mitigates age-related degenerative changes in aged mice.

(A) Changes in the body weights of the mice after enalapril feeding. (B, C) Movement trajectory (B) and spontaneous alternation rate (C) in the Y-maze of the mice after enalapril treatment. (D, E) Changes in the grip strength (D) and rotarod fall time (E) of the mice after enalapril treatment. (F, G) Heatmap of movement trajectories (F) and the number of entries into the central area (G) in the open-field test after enalapril treatment. (H–M) Alterations in the serum levels of aspartate transaminase (AST) (H), creatinine (CREA) (I), lactate dehydrogenase (LDH) (J), triglycerides (TGs) (K), low-density lipoprotein cholesterol (LDL) (L), and high-density lipoprotein cholesterol (HDL) (M) after enalapril treatment in mice. (N) SA- β -Gal staining of the brain, liver, and spleen after enalapril treatment. Scale bars, 20 μ m. (O, P) Periodic Acid-Schiff (PAS) staining (O) and Congo red staining (P) of the brain after enalapril treatment. Scale bars, 50 μ m. (Q) Sirius red staining of the kidney after enalapril treatment. Scale bars, 50 μ m. (R) Oil Red O staining of the spleen, kidney and liver after enalapril treatment. Scale bars, 20 μ m. (S) Hematoxylin and eosin (H&E) staining of the liver after enalapril treatment. Scale bars, 50 μ m.

anxiety (**Fig. 6F, G**). However, LDN treatment reversed the improvements in muscle endurance, spatial memory, and reduced anxiety behaviors conferred by enalapril (**Fig. S8A–F**). In conclusion, these results suggest that enalapril has the potential to mitigate the deterioration of various behavioral aspects associated with aging through the activation of pSmad1/5/9.

Enalapril ameliorates aging-related pathological changes in aged mice

The results of these behavioral tests provided valuable insights. To further explore the effects of enalapril on pathological changes, our investigation was extended to serum analyses. This comprehensive analysis encompassed the evaluation of vital organ function, including the liver, kidneys, and heart, as well as an assessment of lipid content, specifically triglycerides (TGs), low-density lipoprotein cholesterol (LDL), and high-density lipoprotein cholesterol (HDL). After enalapril administration, there was no significant change in the serum alanine transaminase (ALT) level, indicating that no liver toxicity from enalapril occurred (**Fig. 6H**). Moreover, the level of creatinine (CREA), a key marker of kidney function, was significantly decreased in enalapril-treated mice, indicating potential functional recovery and improvement of renal function (**Fig. 6I**). A heart health indicator, lactate dehydrogenase (LDH), was also reduced, indicating a positive cardiac effect of enalapril (**Fig. 6J**). These findings underscore the multifaceted effects of enalapril on various physiological parameters, emphasizing its potential as a therapeutic intervention in aging-related conditions.

The transcriptomic data from the enalapril treatment group revealed an increase in long-chain fatty acid metabolism in the liver, suggesting that enalapril may play a positive role in lipid metabolism (**Fig. S6A**). Accordingly, we analyzed serum lipid levels in enalapril-treated mice. Typically, aging leads to a decline in lipid metabolism in the serum, causing an accumulation of triglycerides (TGs) and low-density lipoprotein (LDL) and a decrease in high-density lipoprotein (HDL), contributing to oxidative stress and conditions such as hyperlipidemia and fatty liver (A. A. Johnson & Stolzinger, 2019). Following enalapril treatment, the mice presented significant reductions in TG and LDL levels and a noticeable increase in HDL levels (**Fig. 6K–M**). These results suggest that enalapril has the potential to increase lipid metabolism in aged mice, thereby mitigating the increase in blood lipid levels.

To evaluate the improvements in the aging process induced by enalapril in various organ systems, we conducted pathological examinations of mouse tissues. A decrease in β -gal staining was observed in the brain, liver and spleen of enalapril-treated mice, suggesting an alleviation of aging-associated phenotypes (**Fig. 6N**, **Fig. S9A–C**). Given the effects of enalapril on brain function, we further examined pathological markers in the brain. Periodic acid-Schiff (PAS) staining, which is indicative of glycogen deposition, revealed a significant reduction in glycogen near the hippocampus (**Fig. 6O**, **Fig. S9D**). Congo red staining, which detects amyloid protein deposition, also revealed reduced staining in the cerebral cortex (**Fig. 6P**, **Fig. S9E**). In the kidney, sirius red staining, which highlights abnormal collagen fibers or fibrosis, revealed significant red staining around blood vessels in control mice. However, this staining was noticeably reduced in the kidneys of the mice treated with enalapril (**Fig. 6Q**, **Fig. S9F**). These findings suggest that enalapril has a beneficial effect on fibrosis in the kidneys of aged mice, which aligns with the observed improvement in plasma creatinine levels (**Fig. 6I**). Furthermore, analysis of spleen, kidney and liver sections from enalapril-treated mice revealed a reduction in Oil Red O staining, with the kidneys and liver showing the most significant changes (**Fig. 6R**, **Fig. S9G–I**). This finding aligned with the consistent decrease in lipid-related molecular content in plasma, indicating that enalapril has a role in reducing age-related lipid accumulation (**Fig. 6K–M**). Hematoxylin and eosin (H&E) staining of the liver revealed a decrease in cavities and increased tissue density in mice treated with enalapril, suggesting its role in enhancing liver cell proliferation and regeneration (**Fig. 6S**, **Fig. S9J**).

In summary, enalapril significantly mitigated multiple pathological markers associated with aging, including lipid droplets, amyloid, glycogen deposition, and fibrosis, in the brain, kidneys, liver, heart and spleen of aged mice. These comprehensive improvements contribute to an overall amelioration of their aging status.

Discussion

Since the early 2000s, the National Institutes of Health (NIH) has been pioneering the Interventions Testing Program (ITP), an initiative aimed at harnessing existing drugs or small molecules to decelerate the aging process (Harrison et al., 2009 [↗](#); Nadon et al., 2008 [↗](#); Strong et al., 2016 [↗](#)). This approach repurposes established medications for novel uses, maximizing the potential of known compounds. It has accelerated the transition of anti-aging therapies from discovery to clinical application, ensures their safety and bolsters a comprehensive anti-aging drug screening strategy.

In our study, we discovered that enalapril, a commonly prescribed ACE inhibitor for hypertension, has significant anti-senescence effects at both the cellular and organismal levels (**Fig. 7** [↗](#)). Enalapril demonstrated a notable ability to attenuate aging-related physiological decline across multiple organs, including the brain, kidneys, and liver. This finding suggests its potential to reduce the risk of disease across multiple organs, slow the overall aging process and extend both the healthspan and lifespan of mice. This observation aligns with previous reports on the lifespan-extending effects of other ACE inhibitors, such as captopril, which have been shown to increase longevity in organisms such as *C. elegans* and mice (Egan et al., 2024 [↗](#); Kumar, Dietrich, & Kornfeld, 2016 [↗](#); Strong et al., 2022 [↗](#)). However, our screening process revealed that enalapril outperformed captopril and other antihypertensive drugs in ameliorating senescent phenotypes, although captopril and other antihypertensive drugs also exhibited the ability to delay senescence. These findings indicate that enalapril may have unique or more potent mechanisms of action that contribute to its anti-senescence effects, warranting further mechanistic studies to better understand its molecular targets.

Mechanistically, our data revealed that enalapril enhances the phosphorylation of Smad1/5/9, a critical signaling pathway that regulates various cellular processes, including cell proliferation and the oxidative stress response. Phosphorylation of Smad1/5/9 facilitates its nuclear translocation, where it acts as a transcriptional regulator to upregulate downstream genes, such as those involved in cell cycle regulation and antioxidative responses. These findings suggest that the ability of enalapril to modulate the pSmad1/5/9 pathway contributes to its protective effects against oxidative stress, a well-established hallmark of aging. By reducing ROS levels and promoting cell proliferation, enalapril effectively delays the progression of cellular senescence and reduces organismal aging-related phenotypes. This finding not only highlights the potential of phosphorylated Smad1/5/9 as a therapeutic target for anti-senescence interventions but also reveals a novel antioxidative mechanism of enalapril action.

Several FDA-approved drugs have been repurposed for anti-aging applications, demonstrating a wide range of mechanisms and effects. These include rapamycin, metformin, and NAD⁺ precursors. For example, rapamycin, which was originally used as an immunosuppressant to prevent organ transplant rejection, inhibits the mTOR pathway, which is associated with cell growth and metabolism (S. C. Johnson, Rabinovitch, & Kaeberlein, 2013 [↗](#)). It extends the lifespan of mice and shows potential anti-aging effects across various species (Harrison et al., 2009 [↗](#)). Compared with other repurposed anti-aging drugs, enalapril offers distinct advantages because of its unique mechanism of action. As an ACE inhibitor, enalapril inherently targets the renin-angiotensin system, providing vascular protection (Schmieder, Hilgers, Schlaich, & Schmidt, 2007 [↗](#); Yamamoto & Rakugi, 2021 [↗](#)). Compared with repurposed drugs such as rapamycin, which may have more pronounced side effects, such as immunosuppression, limiting their long-term use,

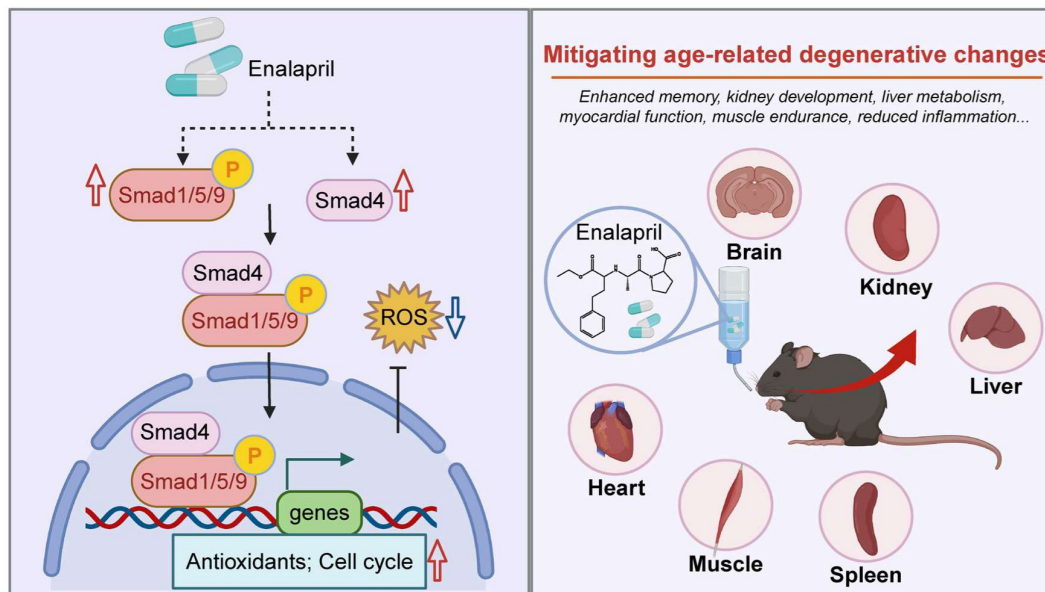


Figure 7

Summary schematic.

It showing pSmad1/5/9-mediated combating cellular senescence of enalapril, and improvement of organ function in mice. Specifically, enalapril increases the level of pSmad1/5/9 and Smad4, promoting the expression of downstream genes, such as antioxidative and cell cycle related genes, thereby promoting cell proliferation and reducing ROS. Moreover, enalapril mitigates age-related degenerative changes, including enhanced memory, improved kidney function, increased liver metabolism and other organ functions. Created with BioRender.com [🔗](#).

enalapril has a well-established safety profile (Janes & Fruman, 2009 [↗](#); Saunders, Metcalfe, & Nicholson, 2001 [↗](#)). This can facilitate faster clinical translation for anti-aging applications. Moreover, our study revealed that enalapril reduces ROS through the pSmad1/5/9-antioxidant axis, highlighting its potential for antioxidation, a hallmark of aging that is less directly targeted by drugs such as rapamycin or metformin. Additionally, enalapril enhances multi-organ function, particularly in the brain, kidneys, and liver. Its well-established safety profile further supports enalapril as a promising candidate for clinical anti-aging applications, with the potential to complement existing therapies by promoting both vascular health and systemic antioxidative and anti-inflammatory effects.

Aging is a highly complex and multifactorial process (Campisi, 2013 [↗](#); Jazwinski, 1996 [↗](#)). It involves intricate interactions among various signaling pathways and crosstalk between tissues and organs, especially during individual aging (Jazwinski, 1996 [↗](#); Munoz-Espin & Serrano, 2014 [↗](#)). Notably, enalapril is not uniform across all organs and tissues, as indicated by the differential improvement in aging phenotypes among various tissues. For example, enalapril significantly improved the physiological functions of the brain, muscle and other organs, but the specific pathways and mechanisms by which these effects are mediated might differ among organs. This observation suggests that the tissue-specific effects of enalapril may be mediated by differential activation of signaling pathways, including those beyond the Smad1/5/9 axis, which are known to play crucial roles in aging. Future studies should focus on delineating these tissue-specific mechanisms to fully understand the scope of the anti-aging effects of enalapril. Moreover, improvements in metabolic function in the liver could have downstream effects on brain function, as metabolic dysregulation is known to exacerbate neurodegenerative diseases. Similarly, enhanced vascular health resulting from the antihypertensive properties of enalapril may contribute to improved brain perfusion, reducing the risk of age-related cognitive decline. Therefore, exploring the systemic effects of enalapril and how improvements in one organ system may benefit others should be a priority for future research. Furthermore, given the established role of enalapril as an antihypertensive drug, assessing its potential side effects when it is repurposed for anti-aging purposes is essential. Identifying strategies to minimize these side effects while optimizing their anti-aging effects is an important area of investigation. In conclusion, our study identified enalapril as a promising candidate for repurposing as an anti-senescence drug with antioxidative and promoting effects on cell proliferation. These findings open avenues for the development of targeted therapies aimed at mitigating aging and promoting healthy longevity in clinical settings, providing a foundation for guiding future research in the realm of anti-aging drug interventions.

Data availability

All related sequencing data have been uploaded to the NCBI Gene Expression Omnibus and are accessible through accession numbers GSE277861 and GSE277862.

Supplementary Tables

Name	Primer sequence
IL1 β	For: ATGATGGCTTATTACAGTGGCAA Rev: GTCGGAGATTCGTAGCTGGA
IL6	For: ACTCACCTCTTCAGAACGAATTG Rev: CCATCTTTGGAAGGTTTCAGGTTG
CXCL10	For: GTGGCATTCAAGGAGTACCTC Rev: TGATGGCCTTCGATTCTGGATT
CXCL16	For: CCCGCCATCGGTTTCAGTTC Rev: CCCCAGAGTAAGCATGTCCAC
CCL2	For: AAGACCATTGTGGCCAAGGA Rev: TTCGGAGTTTGGGTTTGCT
MMP2	For: CCCACTGCGGTTTTCTCGAAT Rev: CAAAGGGGTATCCATCGCCAT
BMPRI1A	For: CTTTACCACTGAAGAAGCCAGCT Rev: AGAGCTGAGTCCAGGAACCTGT
ID1	For: CTGCTCTACGACATGAACGG Rev: GAAGGTCCCTGATGTAGTCGAT
ID2	For: AGTCCCGTGAGGTCCGTTAG Rev: AGTCGTTTCATGTTGTATAGCAGG
ID3	For: GAGAGGCACTCAGCTTAGCC Rev: TCCTTTTGTTCGTTGGAGATGAC
ACTB	For: CACCCCGTGCTGCTGAC Rev: CCAGAGGCGTACAGGGATAG
TXN	For: GTGAAGCAGATCGAGAGCAAG Rev: CGTGGCTGAGAAGTCAACTACTA
GPX4	For: GAGGCAAGACCGAAGTAACTAC Rev: CCGAACTGGTTACACGGGAA
PRDX5	For: CTTACCCCTGGATGTTCCAA Rev: AGGCATCATTAACACTCAGACAG

Table S1

Primers used in RT-qPCR

Name	Primer sequence	Notes
HPRT1	For: TGTTTGGGCTATTTACTAGTTG Rev: ATAAAATGACTTAAGCCCAGAG	Negative control (Morikawa et al., 2011; Sullivan & Santos, 2020)
HBB	For: GGGCTGAGGGTTTGAAGTCC Rev: CATGGTGTCTGTTTGAGGTTGC	Negative control (Morikawa et al., 2011)
ID1	For: AGTCCGTCGGGGTTTTATG Rev: TGTGTCAGCGTCTGAACCAG	Positive control (Morikawa et al., 2011; Sullivan & Santos, 2020)
ID2	For: ACTCTATTTACCACCCCAGC Rev: AGCTTCCCTTCGTCCCCAT	
TXN	For: CAGGGCTGGATTCCTCGAAA Rev: CAAGGACGTACACACCGAGA	
PRDX5	For: GTATGGGACTAGCTGGCGTG Rev: TCACTGTACCGTCTTGCTGC	
GPX4	For: GAAGCAGAGACGGGAGGTTC Rev: CTTGTGTCTAGGAGGCCGTG	
SOD3	For: TCTGAGGGGTTAGTGGGGAG Rev: CCCCTTTCTCGTCACTCCAG	

Table S2

Primers used in pSmad1/5/9 ChIP-qPCR

Name	Target Sequence
shID1 #1	CCGGCCTACTAGTCACCAGAGACTTCTCG AGAAGTCTCTGGTGACTAGTAGGTTTT
shID1 #2	CCGGACTCGGAATCCGAAGTTGGAAGTC GAGTTCCAAGTTCGGATTCCGAGTTTTT
shID2 #1	CCGGCCCTTCTGAGTTAATGTCAAAGTCG AGTTTGACATTAAGTCAGAAGGGTTTT
shID2 #2	CCGGCCCACTATTGTCAGCCTGCATCTCG AGATGCAGGCTGACAATAGTGGGTTTT
shBMPRI1A	CCGGCGCCAATCTCATACAAGCCATCTC GAGATGGCTTGTATGAGATTGGCGTTTT

Table S3

shRNA target sequence used in knock down experiment

Name	Source	Number
Mouse monoclonal anti- β -actin	Santa Cruz Biotechnology	Cat#sc-47778
Rabbit monoclonal anti-CDKN2A/p16INK4a	Abcam	Cat#ab108349
Rabbit monoclonal anti-p21	Abcam	Cat#ab109199
Rabbit monoclonal anti-Phospho-SMAD1 (Ser463/465)/ SMAD5 (Ser463/465)/ SMAD9 (Ser465/467)	Cell Signaling Technology	Cat#13820S
Rabbit monoclonal anti-SMAD1 + SMAD5 + SMAD9 (phosphor S463 + S465 + S467)	Abcam	Cat#ab92698
Rabbit monoclonal anti-Phospho-SMAD5 (Ser463/465)	Huabio	Cat#ET1605-5
Rabbit polyclonal anti-SMAD1/5/9	Immunoway	Cat#YT4325
Rabbit monoclonal anti-SMAD5	Huabio	Cat#ET1606-26
Rabbit monoclonal anti-SMAD4	Abcam	Cat#ab40759
Rabbit polyclonal anti-SMAD2 (Ser465/467)	Cell Signaling Technology	Cat#3101S
Rabbit monoclonal anti-SMAD3 (Ser423/425)	Cell Signaling Technology	Cat#9520S
Rabbit polyclonal anti-SMAD2/3	Cell Signaling Technology	Cat#3102S
Rabbit polyclonal anti-BMP2	Huabio	Cat#ER80602
Mouse monoclonal anti-BMP4	Huabio	Cat#EM1706-23
Rabbit polyclonal anti-ID1	Immunoway	Cat#YN0093
Rabbit polyclonal anti-ID2	Immunoway	Cat#YN2457
Rabbit polyclonal anti-PRDX5	ABclonal	Cat#A1269
Rabbit monoclonal anti-TXN	ABclonal	Cat#A4024
Rabbit monoclonal anti-TXN2	ABclonal	Cat#A4424
Rabbit polyclonal anti-NDUFB10	ABclonal	Cat#A9382
Rabbit monoclonal anti-GPX4	Abmart	Cat#T56959
IRDy 800CW Goat anti-Mouse IgG Secondary Antibody	LI-COR	Cat#926-32210
IRDy 800CW Goat anti-Rabbit IgG Secondary Antibody	LI-COR	Cat#926-32211

Table S4

Antibodies

Supplementary Figures and legends

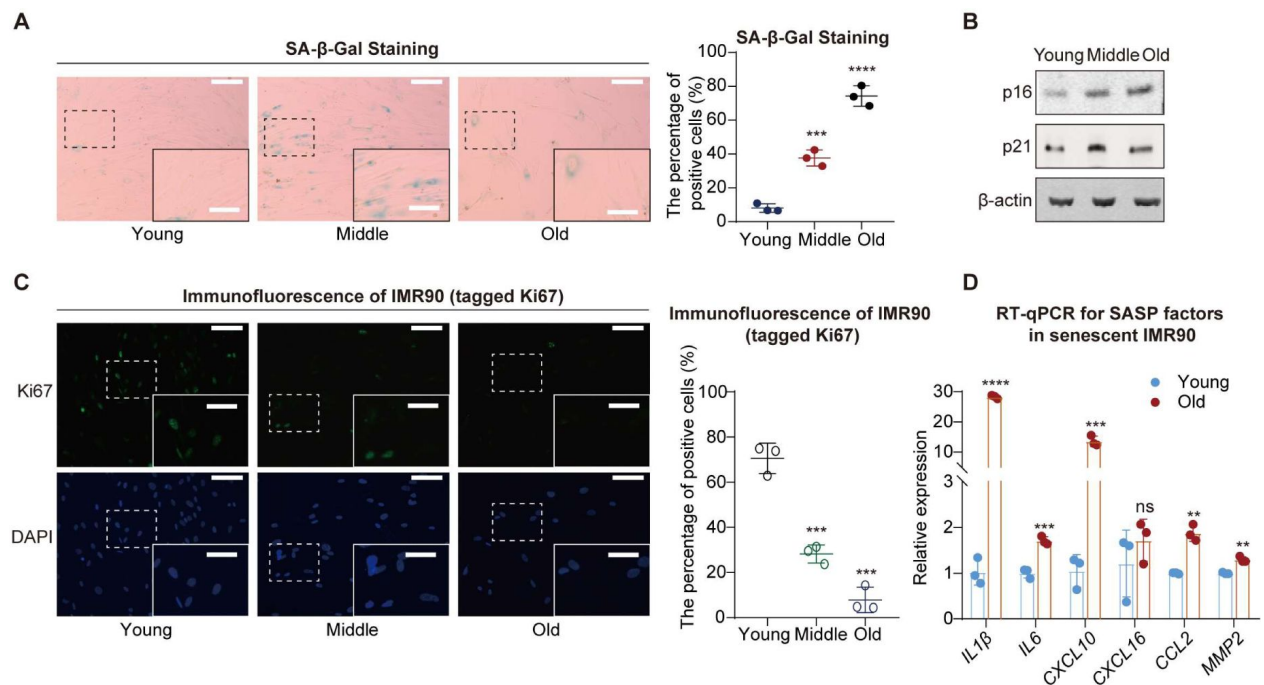


Figure S1

The expression of various senescence markers during IMR90 cell senescence.

(A) SA-β-Gal staining (left) and statistical analysis of SA-β-Gal ratios (right) during IMR90 cell senescence. Scale bars, 200 μm. Enlarged scale bars, 100 μm. (B) Western blot analysis showing the protein levels of p16 and p21 in IMR90 cells at different passage. (C) Ki67 immunofluorescence experiment (left) and statistical analysis of Ki67 intensity (right) during IMR90 cell senescence. Scale bars, 80 μm. Enlarged scale bars, 40 μm. (D) RNA expression of SASP factors during IMR90 cell senescence, with blue bars representing the young group and red bars representing the old group. A two-tailed t-test was employed, ns indicates no significant difference, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Figure S2

Alterations in critical regulators of cellular senescence.

(A) Western blot analysis showing the changes in protein levels of key targets in classical senescence signaling pathways following enalapril treatment. (B) Western blot analysis showing the changes in Smad and BMP-related protein levels during IMR90 cell senescence.

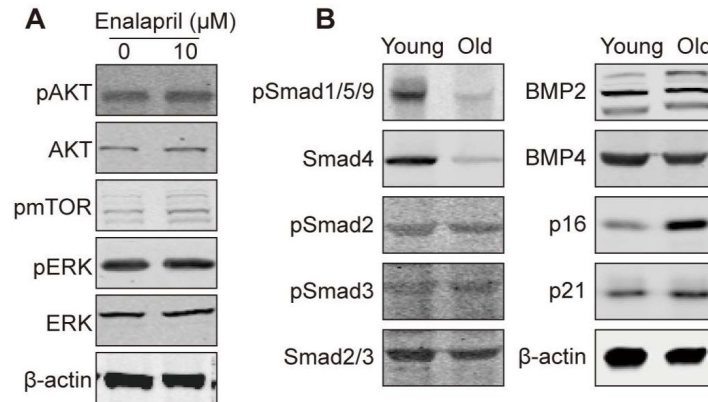
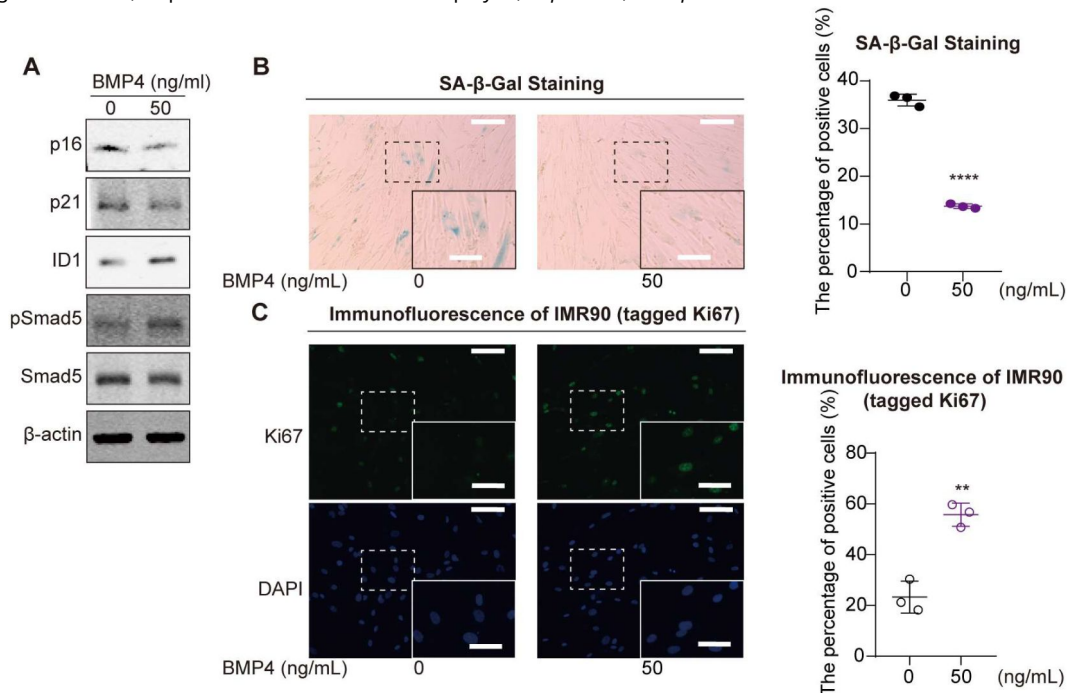


Figure S3

pSmad1/5/9 alleviates cellular senescence phenotypes.

(A) Western blot analysis showing the changes in protein levels following BMP4 treatment. (B) SA- β -Gal staining (left) and corresponding ratio analysis chart (right) following BMP4 treatment. Scale bars, 200 μ m. Enlarged scale bars, 100 μ m. (C) Ki67 immunofluorescence (right) and corresponding ratio analysis chart (left) following BMP4 treatment. Scale bars, 80 μ m. Enlarged scale bars, 40 μ m. A two-tailed t-test was employed, ** $p < 0.01$, **** $p < 0.0001$.



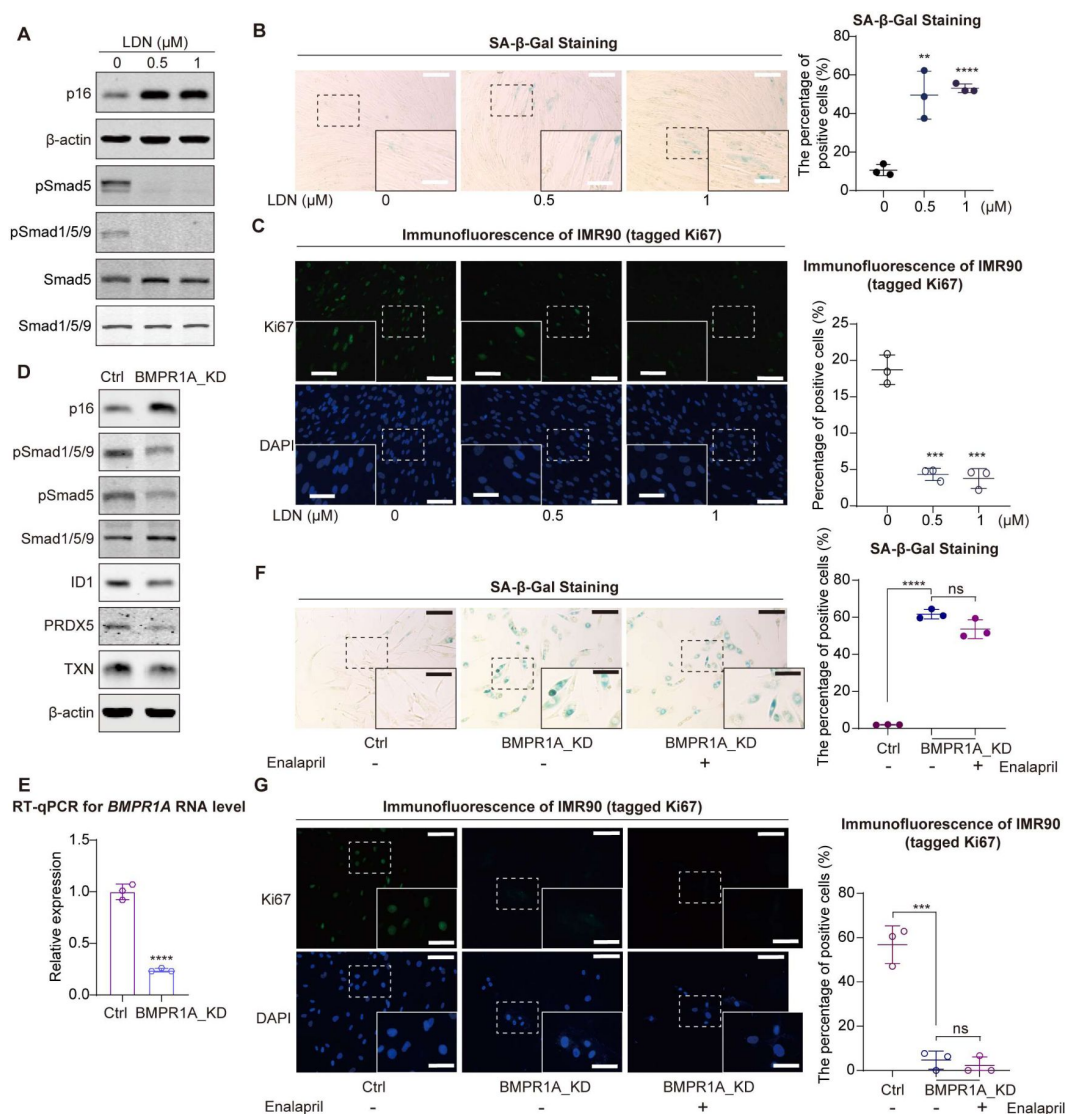


Figure S4

Reduction of pSmad1/5/9 correlates with cellular senescence.

(A) Western blot showing the protein levels after the addition of LDN193189. (B) SA- β -gal staining (left) after adding LDN193189 and its corresponding staining proportion statistical chart (right). Scale bars, 200 μm . Enlarged scale bars, 100 μm . (C) Immunofluorescence staining for Ki67 (left) after adding LDN193189 along with the corresponding intensity statistical chart (right). Scale bars, 80 μm . Enlarged scale bars, 40 μm . (D) Western blot showing the protein levels following BMPR1A knockdown. (E) Relative RNA levels of BMPR1A following BMPR1A knockdown. (F) SA- β -gal staining (left) and its corresponding staining proportion statistical chart (right) in the control group (Ctrl) and *BMPR1A*-knockdown groups (*BMPR1A_KD*) with enalapril treatment. Scale bars, 200 μm . Enlarged scale bars, 100 μm . (G) Ki67 immunofluorescence (left) and its corresponding staining proportion statistical chart (right) in the control group (Ctrl) and *BMPR1A*-knockdown groups (*BMPR1A_KD*) with enalapril treatment. Scale bars, 80 μm . Enlarged scale bars, 40 μm . A two-tailed t-test was employed, ns indicates no significant difference, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$.

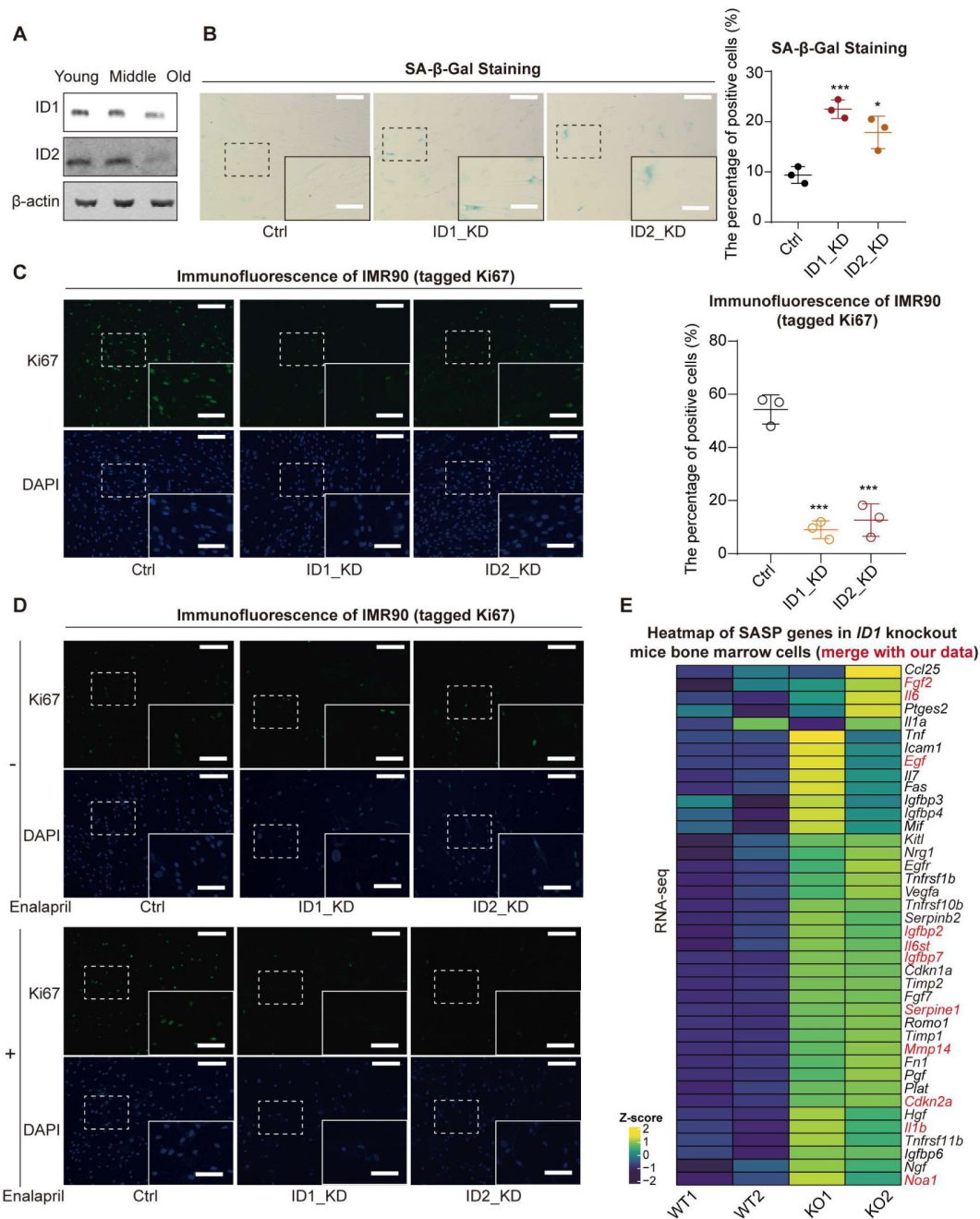


Figure S5

Knockdown of *ID* accelerates cellular senescence.

(A) Western blot showing changes in ID protein levels during cellular senescence. (B) SA-β-Gal staining (left) following knockdown of *ID1* and *ID2*, along with corresponding statistical chart (right). Scale bars, 200 μm. Enlarged scale bars, 100 μm. (C) Ki67 immunofluorescence (left) following knockdown of *ID1* and *ID2*, along with corresponding statistical chart (right). Scale bars, 80 μm. Enlarged scale bars, 40 μm. A two-tailed t-test was employed, * $p < 0.05$, *** $p < 0.001$. (D) Ki67 immunofluorescence in control group (Ctrl) and *ID* knockdown groups (*ID1_KD*, *ID2_KD*), with or without enalapril treatment. Scale bars, 80 μm. Enlarged scale bars, 40 μm. (E) Heatmap of SASP factor expression in mouse bone marrow cells following *ID* knockout from previously reported data (Fei et al., 2023).

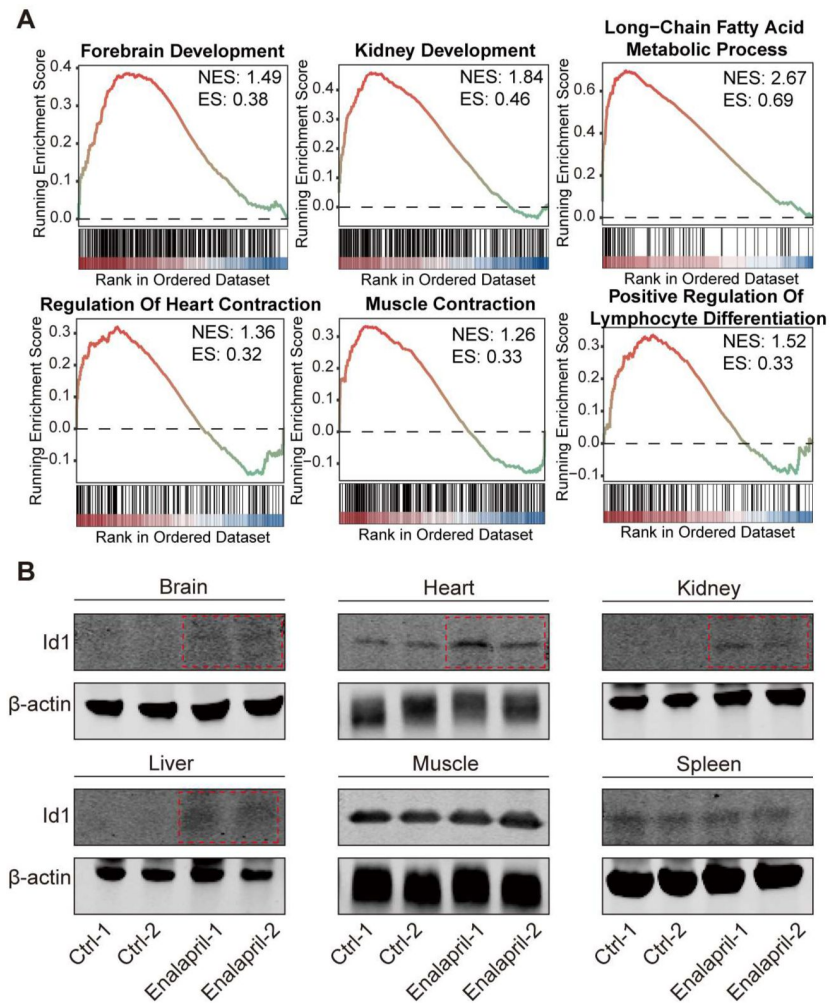


Figure S6

Effects of enalapril on the expression changes of genes related to organ physiological functions in mice.

(A) In multiple organs, enalapril increases the enrichment of GSEA pathways associated with improved physiological function.
 (B) Western blot analysis showing changes in Id1 protein levels in various organs of mice.

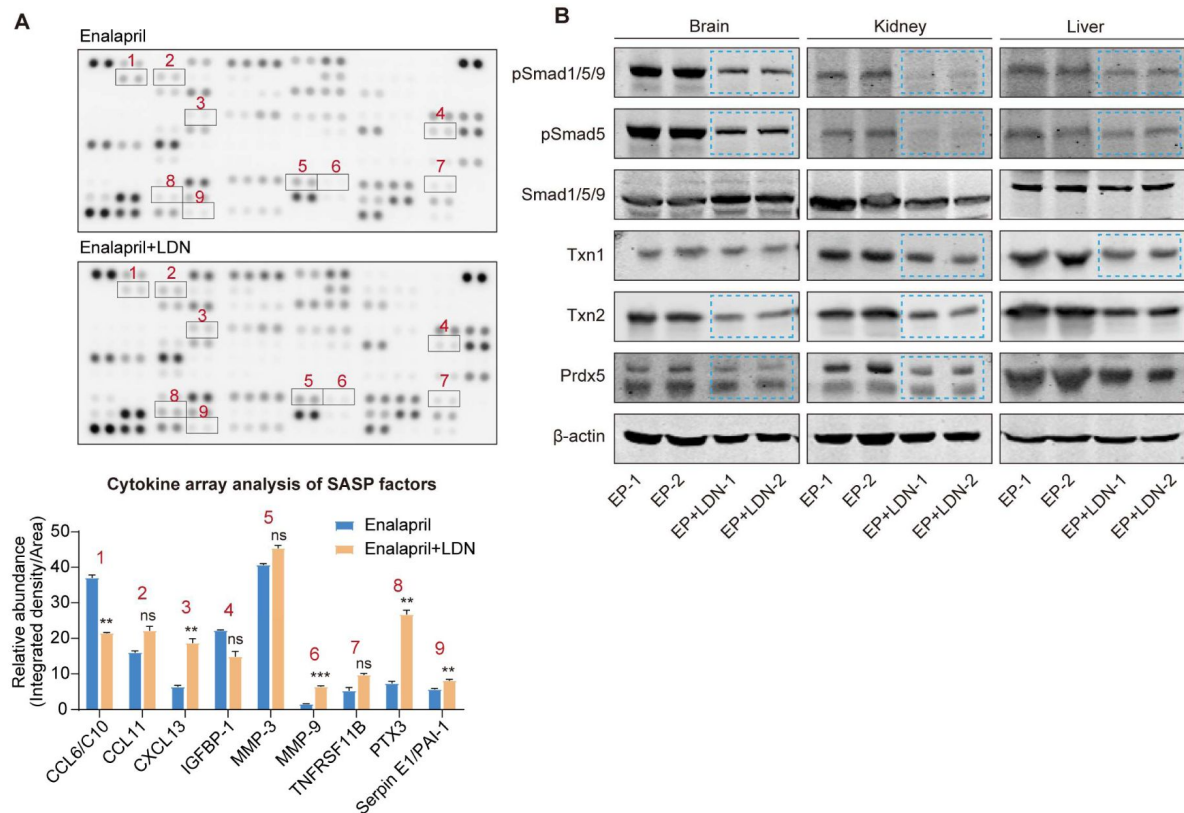


Figure S7

Effects of reduced pSmad1/5/9 levels on the expression changes of genes in mice.

(A) Cytokine array analysis of secreted proteins (above) and relative quantitation (below) of SASP factors in enalapril-treated serum (Enalapril) and co-treated serum (Enalapril+LDN). (B) Western blot analysis showing the protein levels of pSmad1/5/9 and antioxidative genes in various organs following enalapril and LDN co-treatment.

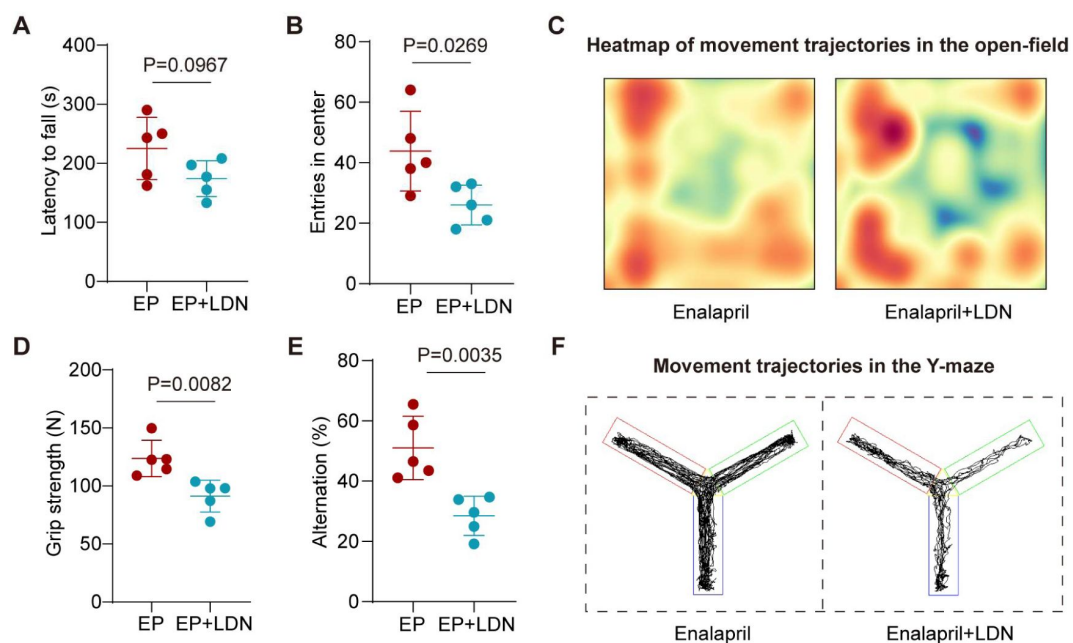


Figure S8

Effects of reduced pSmad1/5/9 levels on aging-related behaviors in mice.

(A, D) Changes in the rotarod fall time (A) and grip strength (D) of the mice after enalapril and LDN treatment. (B, C) The number of entries into the central area (B) and heatmap of movement trajectories (C) in the open-field test after enalapril and LDN treatment. (E, F) Spontaneous alternation rate (E) and movement trajectory (C) in the Y-maze of the mice after enalapril and LDN treatment.

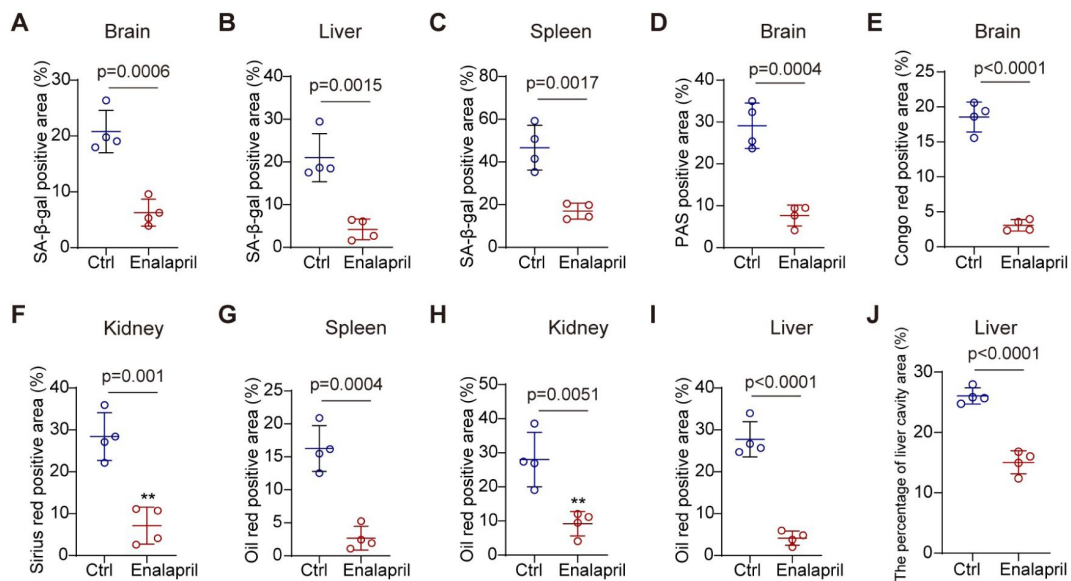


Figure S9

Enalapril ameliorates pathological phenotypes in aged mice.

(A-C) SA- β -Gal staining quantification chart of the brain (A), liver (B) and spleen (C) after enalapril treatment. (D, E) Periodic Acid-Schiff (PAS) staining (D) and Congo red staining (E) quantification chart of the brain after enalapril treatment. (F) Sirius red staining quantification chart of the kidney after enalapril treatment. (G-I) Oil Red O staining quantification chart of the spleen (G), kidney (H) and liver (I) after enalapril treatment. (J) Hematoxylin and eosin (H&E) staining quantification chart of the liver after enalapril treatment.

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Additional information

Author contributions

W.T. oversaw the study and obtained financial support. W.C.L. conducted the main bench experiments. H.C.W. analyzed the data. Z.H.D. and J.N.H. helped prepare the mice samples. R.W., F.J.M. and J.L.B. helped analyze the data. L.J.Z. helped guide the experiment. W.C.L., Y. T. G. and C. Z. prepared the original manuscripts. All the authors read and approved the final version.

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Additional files

supplementary information 

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Reviewer #1 (Public review):

Summary:

In this study, the authors showed that enalapril was able to reduce cellular senescence and improve health status in aged mice. The authors further showed that phosphorylated Smad1/5/9 was significantly elevated and blocking this pathway attenuated the protection of cells from senescence. When middle-aged mice were treated with enalapril, the physiological performance in several tissues, including memory capacity, renal function and muscle strength, exhibited significant improvement.

Strengths:

The strength of the study lies in the identification of pSMAD1/5/9 pathway as the underlying mechanism mediating the anti-senescence effects of enalapril with comprehensive evaluation both in vitro and in vivo.

Weaknesses:

The major weakness of the study is the in vivo data. Despite the evidence shown in the in vitro study, there is no data to show that blocking the pSmad1/5/9 pathway is able to attenuate the anti-aging effects of enalapril in the mice. In addition, the aging phenotypes mitigation by enalapril is not evidenced by the extension of lifespan. If it is necessary to show that NAC is able to attenuate enalapril effects in the aging mice. In addition, it would be beneficial to test if enalapril is able to achieve similar rescue in a premature aging mouse model.

Comments on revisions:

The revised manuscript provided additional in vivo data that addressed my questions accordingly. I think the authors have done an excellent job in demonstrating that enalapril improved physiological phenotypes in aged mice through pSmad1/5/9 pathway.

Their response to my question regarding the test in HGPS mice was not satisfactory. Premature aging and physiological aging share substantial similarities in their pathways. Given that this is not the focus of current study and the manuscript does not provide data on HGPS mice, I think this does not affect the conclusion of the current study.

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Reviewer #2 (Public review):

This manuscript presents an interesting study of enalapril for its potential impact on senescence through the activation of Smad1/5/9 signaling with a focus on antioxidative gene expression. Repurposing enalapril in this context provides a fresh perspective on its effects beyond blood pressure regulation. The authors make a strong case for the importance of Smad1/5/9 in this process, and the inclusion of both in vitro and in vivo models adds value to the findings. Below, I have a few comments and suggestions which may help improve the manuscript.

A major finding in the study is that phosphorylated Smad1/5/9 mediates the effects of enalapril. However, the manuscript focused on the Smad pathway relatively abruptly, and

the rationale behind targeting this specific pathway is not fully explained. What makes Smad1/5/9 particularly relevant to the context of this study?

Furthermore, their finding that activation of Smad1/5/9 leads to a reduction of senescence appears somewhat contradictory to the established literature on Smad1/5/9 in senescence. For instance, studies have shown that BMP4-induced senescence involves activation of Smad1/5/8 (Smad1/5/9), leading to the upregulation of senescence markers like p16 and p21 (JBC, 2009, 284, 12153). Similarly, phosphorylated Smad1/5/8 has been shown to promote and maintain senescence in Ras-activated cells (PLOS Genetics, 2011, 7, e1002359). Could the authors provide more detailed mechanistic insights into why enalapril seems to reverse the typical pro-senescent role of Smad1/5/9 in their study?

While the authors showed that enalapril increases pSmad1/5/9 phosphorylation, what are the expression levels of other key and related factors like Smad4, pSmad2, pSmad3, BMP2, and BMP4 in both senescent and non-senescent cells? These data will help clarify the broader signaling effects.

They used BMP receptor inhibitor LDN193189 to pharmacologically inhibit BMP signaling, but it would be more convincing to also include genetic validation (e.g., knockdown or knockout of BMP2 or BMP4). This will help confirm that the observed effects are truly due to BMP-Smad signaling and not off-target effects of the pharmacological inhibitor LDN.

I don't see the results on the changes in senescence markers p16 and p21 in the mouse models treated with enalapril. Similarly, the effects of enalapril treatment on some key SASP factors, such as TNF- α , MCP-1, IL-1 β , and IL-1 α , are missing, particularly in serum and tissues. These are important data to evaluate the effect of enalapril on senescence.

Given that enalapril is primarily known as an antihypertensive, it would be helpful to include data on how it affects blood pressure in the aged mouse models, such as systolic and diastolic blood pressure. This will clarify whether the observed effects are independent of or influenced by changes in blood pressure.

<https://doi.org/10.7554/eLife.104774.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

In this study, the authors showed that enalapril was able to reduce cellular senescence and improve health status in aged mice. The authors further showed that phosphorylated Smad1/5/9 was significantly elevated and blocking this pathway attenuated the protection of cells from senescence. When middle-aged mice were treated with enalapril, the physiological performance in several tissues, including memory capacity, renal function, and muscle strength, exhibited significant improvement.

Strengths:

The strength of the study lies in the identification of the pSMAD1/5/9 pathway as the underlying mechanism mediating the anti-senescence effects of enalapril with comprehensive evaluation both in vitro and in vivo.

Thank you for your patient reading and great efforts to advance our research! Your comments are shown in bold font below, and specific concerns have been numbered. Our point-by-point answers are provided in standard blue font, with all modifications and additions to the MS highlighted in red text.

Weaknesses:

(1) The major weakness of the study is the in vivo data. Despite the evidence shown in the in vitro study, there is no data to show that blocking the pSmad1/5/9 pathway is able to attenuate the anti-aging effects of enalapril in the mice. In addition, the aging phenotypes mitigation by enalapril is not evidenced by the extension of lifespan.

Many thanks for your careful reading and valuable comments! We fully agree with this comment. In accordance with your suggestion, we administered LDN193189 to investigate its suppressive effects on pSmad1/5/9 signaling in vivo. Notably, pharmacological inhibition of pSmad1/5/9 resulted in upregulation of enalapril-suppressed SASP factors, while conversely leading to marked decrease of downstream antioxidant genes expression across multiple organ systems (Revised Fig. S7). These analyses and corresponding sentences have been added in the Result section of the revised MS (Revised Fig.S7, Lines 222–223, 444–448).

Additionally, aging-related behavioral phenotypes were also examined following pSmad1/5/9 inhibition, including decreased muscle strength and endurance, impaired spatial memory and increased anxiety behaviors (Revised Fig. S8). These analyses and corresponding sentences have been added in the Result section of the revised MS (Revised Fig.S8, Lines 476–480). Collectively, these findings demonstrate that the anti-aging effects of enalapril in mice are mediated through the pSmad1/5/9 pathway.

In this study, we focused exclusively on assessing the improvement in the health status of aged mice, which indicates that enalapril can extend the healthspan of aged mice. While we agree that lifespan extension is an important indicator of anti-aging potential, recent studies have emphasized that healthspan, rather than lifespan alone, provides a more relevant and translational measure of aging interventions, particularly in the context of chronic disease and quality of life in aged individuals (Kennedy et al., 2014; Lopez-Otin et al., 2023). Moreover, given the strong influence of genetic background, environmental factors and stochastic events on lifespan, focusing on functional rejuvenation and delayed onset of aging-related pathologies may offer a more practical and mechanistically informative approach. Our study aims to elucidate how enalapril enhances healthy phenotypes in aged mice, however, we acknowledge the critical need for direct lifespan evaluation and intend to address this limitation in subsequent research. We sincerely hope that these explanations address your concerns.

(2) If it is necessary to show that NAC is able to attenuate enalapril effects in the aging mice. In addition, it would be beneficial to test if enalapril is able to achieve similar rescue in a premature aging mouse model.

Thanks for your suggestion. We apologize for any confusion that may have arisen due to the wording in the original manuscript. N-acetylcysteine (NAC) is widely reported as an antioxidant that scavenges reactive oxygen species (ROS) (Huang et al., 2020; Zafarullah et al., 2003). In our study, enalapril was also observed to reduce ROS levels. Therefore, NAC is unlikely to antagonize the effects of enalapril in this context, as both compounds act in a similar direction with respect to oxidative stress mitigation. To avoid potential misunderstanding, we have carefully reviewed the relevant statements in the MS and revised the text to clarify this point.

We sincerely appreciate this valuable suggestion to evaluate enalapril in a premature aging mouse model; however, the premature aging mouse models represent a pathological form of aging, whereas the naturally aged mouse models used in our study reflect physiological aging processes. While we observed beneficial effects of enalapril in naturally aged mice, these effects may not necessarily extend to premature aging models due to fundamental differences in the underlying mechanisms and progression of aging. Natural aging is characterized by the gradual accumulation of cellular damage, driven by multifactorial processes such as inflammaging and mitochondrial dysfunction. In this context, enalapril appears effective, in part by modulating SASP factors and reducing oxidative stress through the BMP-Smad signaling axis (Revised Fig. 4, 5) (Lopez-Otin et al., 2023). In contrast, premature aging models are driven by distinct mechanisms like nuclear lamina defects, which may not respond similarly to BMP-Smad axis. Moreover, genetic background, strain variability, and specific model characteristics can significantly influence treatment outcomes (Mitchell et al., 2016). For instance, rapamycin extends lifespan in wild-type mice but shows limited effects on aging, underscoring the challenge of extrapolating findings across distinct aging models (Neff et al., 2013). We sincerely hope that these explanations address your concerns. Thank you again for your great efforts in advancing our research!

Reviewer #2 (Public review):

This manuscript presents an interesting study of enalapril for its potential impact on senescence through the activation of Smad1/5/9 signaling with a focus on antioxidative gene expression. Repurposing enalapril in this context provides a fresh perspective on its effects beyond blood pressure regulation. The authors make a strong case for the importance of Smad1/5/9 in this process, and the inclusion of both in vitro and in vivo models adds value to the findings. Below, I have a few comments and suggestions which may help improve the manuscript.

We appreciate your great efforts in advancing our research! Your comments are shown in bold font below, and specific concerns have been numbered. Our point-by-point answers are provided in standard blue font, with all modifications and additions to the MS highlighted in red text.

(1) A major finding in the study is that phosphorylated Smad1/5/9 mediates the effects of enalapril. However, the manuscript focused on the Smad pathway relatively abruptly, and the rationale behind targeting this specific pathway is not fully explained. What makes Smad1/5/9 particularly relevant to the context of this study?

Thank you for your informative guidance, and we regret for the unclear description. As stated in the MS, after we found that enalapril could improve the cellular senescence phenotype, we screened and examined key targets in important aging-related signaling pathways, such as AKT, mTOR, ERK, Smad2/3 and Smad1/5/9 (Revised Fig. S2A, Revised Fig. 2A). We found that only the phosphorylation levels of Smad1/5/9 significantly increased after enalapril treatment. Therefore, the subsequent focus of this study is on pSmad1/5/9. We sincerely hope that these explanations address your concerns.

(2) Furthermore, their finding that activation of Smad1/5/9 leads to a reduction of senescence appears somewhat contradictory to the established literature on Smad1/5/9 in senescence. For instance, studies have shown that BMP4-induced senescence involves the activation of Smad1/5/8 (Smad1/5/9), leading to the upregulation of senescence markers like p16 and p21 (JBC, 2009, 284, 12153). Similarly, phosphorylated Smad1/5/8 has been shown to promote and maintain senescence in Ras-activated cells (PLOS Genetics, 2011, 7, e1002359). Could the authors provide more detailed mechanistic

insights into why enalapril seems to reverse the typical pro-senescent role of Smad1/5/9 in their study?

Many thanks for your helpful comments! The downstream regulatory network of BMP-pSmad1/5/9 is highly complex. The BMP-SMAD-ID axis has been mentioned in many studies, and its downstream signaling inhibits the expression of p16 and p21 (Hayashi et al., 2016; Ying et al., 2003). Additionally, studies have also found that the Smad1-Stat1-P21 axis inhibits osteoblast senescence (Xu et al., 2022). In our study, enalapril was found to increase the expression of ID1, which is a classic downstream target of pSmad1/5/9 (Genander et al., 2014). Therefore, pSmad1/5/9 inhibits cellular senescence markers such as p16, p21 and SASP through ID1, thereby promoting cell proliferation (Revised Fig. 3). Furthermore, we also found that pSmad1/5/9 increases the expression of antioxidant genes and reduces ROS levels, exerting antioxidant effects (Revised Fig. 4). Together, ID1 and antioxidant genes enable pSmad1/5/9 to exert its anti-senescence effects. We sincerely hope that these explanations address your concerns.

(3) While the authors showed that enalapril increases pSmad1/5/9 phosphorylation, what are the expression levels of other key and related factors like Smad4, pSmad2, pSmad3, BMP2, and BMP4 in both senescent and non-senescent cells? These data will help clarify the broader signaling effects.

Thanks for your insightful suggestions. We observed an increase in pSmad1/5/9 and Smad4 expression, while the levels of pSmad2 and pSmad3 remained unchanged after enalapril treatment (Revised Fig. 2A). Consistently, we found that the levels of pSmad1/5/9 and Smad4 were markedly reduced in senescent cells, aligning with the upregulation of these proteins by enalapril (Revised Fig. S2B). In contrast, pSmad2 and pSmad3 showed a slight increase during senescence, while BMP2 and BMP4 were slightly decreased, though these changes were not statistically significant (Revised Fig. S2B). These findings suggest that enalapril primarily exerts its effects by enhancing pSmad1/5/9 and Smad4 levels, thereby regulating downstream target genes and contributing to the restoration of a more youthful cellular state. These analyses and corresponding sentences have been added in the Result section of the revised MS (Revised Fig.S2B, Lines 303–306, 311–313).

(4) They used BMP receptor inhibitor LDN193189 to pharmacologically inhibit BMP signaling, but it would be more convincing to also include genetic validation (e.g., knockdown or knockout of BMP2 or BMP4). This will help confirm that the observed effects are truly due to BMP-Smad signaling and not off-target effects of the pharmacological inhibitor LDN.

Many thanks for your careful reading and valuable comments! We used shRNA to knockdown the BMP receptor BMPR1A, which led to a reduction in Smad1/5/9 phosphorylation (Revised Fig. S4D, E). This was accompanied by senescence-associated phenotypes, including increased expression of p16 and SA- β -gal and decreased Ki67 staining (Revised Fig. S4F, G). Notably, the addition of enalapril failed to reverse these senescence phenotypes under BMPR1A knockdown conditions, mirroring the results observed with the BMP receptor inhibitor LDN193189 (Revised Fig. S4F, G, Revised Fig. 2F, G). Furthermore, knockdown of BMPR1A also resulted in a marked decrease in the expression of downstream targets, such as ID1 and antioxidative genes (Revised Fig. S4D). These findings strongly support the notion that enalapril exerts its anti-senescence effects through BMP-Smad signaling. These analyses and corresponding sentences have been added in the Result section of the revised MS (Revised Fig.S4D–G, Lines 323–329, 335–337, 348–351, 416–418).

(5) I don't see the results on the changes in senescence markers p16 and p21 in the mouse models treated with enalapril. Similarly, the effects of enalapril treatment on

some key SASP factors, such as TNF- α , MCP-1, IL-1 β , and IL-1 α , are missing, particularly in serum and tissues. These are important data to evaluate the effect of enalapril on senescence.

Thanks for your comments. As for the markers p16 and p21, we observed no change in p16, while the changes in p21 varied across different organs and tissues. Nevertheless, behavioral experiments and physiological and biochemical indicators at the individual level consistently demonstrated the significant anti-aging effects of enalapril (Revised Fig. 6).

We also examined the changes in SASP factors in the serum of mice after enalapril treatment. Notably, SASP factors such as CCL (MCP), CXCL and TNFRS11B showed significant decreases (Revised Fig. 5C). The expression changes of SASP factors varied across different organs. In the liver, kidneys and spleen, the expression of IL1 α and IL1 β decreased, while TNFRS11B expression decreased in both the liver and muscles (Revised Fig. 5B). Additionally, CCL (MCP) levels decreased in all organs (Revised Fig. 5B). We sincerely hope that these explanations address your concerns.

(6) Given that enalapril is primarily known as an antihypertensive, it would be helpful to include data on how it affects blood pressure in the aged mouse models, such as systolic and diastolic blood pressure. This will clarify whether the observed effects are independent of or influenced by changes in blood pressure.

Thanks for your comments. While enalapril is primarily recognized for its antihypertensive properties, in our experimental setting involving aged, normotensive mice, we did not observe notable changes in systolic or diastolic blood pressure following enalapril administration. This observation aligns with previous reports indicating that enalapril does not significantly affect blood pressure in similar non-hypertensive aging models (Keller et al., 2019). Based on these findings, we cautiously interpret that the beneficial effects of enalapril observed in our study are unlikely to be driven by changes in blood pressure. We sincerely hope that these explanations address your concerns. Again, thank you for the constructive comments to advance the understanding of our work!

Reviewer #1 (Recommendations for the authors):

This is an interesting study that reveals enalapril is able to elevate the pSmad1/5/9 pathway to reduce ROS and inflammation to improve the health status in vitro and in vivo. While the pathway is clearly shown in cells to be involved in the enalapril-mediated mitigation of aging, little was done to demonstrate this pathway is responsible for the in vivo effects in the physiological improvements. This can be done by ROS-reduction chemicals such as NAC and also the use of BMP receptor inhibitor LDN193189 (LDN). It is critical to show the lifespan extension in enalapril-treated animals given that the significantly improved physiological functions.

Thanks very much for your constructive recommendations. This part has already been addressed in our response to the public review.

Reviewer #2 (Recommendations for the authors):

The term "anti-aging" appears frequently throughout the manuscript, including in the title. However, the study doesn't directly address lifespan or a comprehensive range of aging symptoms, which are also difficult to define and measure. Many of the observed effects appeared to be driven by senescence. To be more accurate, I recommend avoiding terms like "anti-aging" and "mitigates aging", and instead replacing them with more specific phrases such as "anti-senescence", "senescence reduction/suppression", or

"mitigates age-related symptoms" to better reflect the scope of the study and avoid overstating the findings.

Thanks very much for your constructive recommendations. In accordance with your suggestion, we have revised all uses of the term “aging” in the MS. To facilitate review, all changes have been clearly marked in red text.

Please provide detailed information on the antibodies used, particularly those targeting pSmad1/5/9 and other Smads.

Thanks for your helpful comment. In response, we have now provided detailed information regarding the antibodies used in this study in Revised Table S4 (Revised MS, Page 120–121).

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